Fpt University

TRƯỜNG ĐẠI HỌC FPT;

CAPSTONE PROJECT REPORT

Report 3 – Software Requirement Specification

– Ho Chi Minh, October 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
14/09/26	A	ThongLQ	Added AI-assisted assessment, context and workflow diagrams, and detailed functional requirements.
16/09/26	M	Project team	Updated ownership of the four mainflows: NghiaHQ — MF1; KhanhGNG — MF2; DucDDA — MF3; ThongLQ — MF4.
19 October 2026	M	NghiaHQ	Revised mission planning to use one current mission and reviewed update requests; retained validation and decision evidence.
20 October 2026	M	Project team	Completed functional requirements, lifecycle and authorization rules, pilot account/license workflows, telemetry/report policies and historical similarity evidence; corrected incomplete wording and traceability.

A = Added; M = Modified; D = Deleted.

II. Software Requirement Specification

1. Product Overview

DroneOps Intelligence manages UAVs, pilot accounts and flight credentials, mission planning and reviewed updates, configured flight restrictions, live telemetry, incidents, post-flight evidence and historical-mission-based decision support. The Web application supports Operators and Operations Managers. The Operator mobile client covers account/profile access, own/assigned mission viewing, validation and decision results, live monitoring, alerts and history; mission route editing and Manager administration are Web functions.

The platform validates routes against configured restrictions before Manager approval. A MAVLink-compatible UAV/autopilot or simulator supplies telemetry through an integration gateway. The platform detects configured abnormal conditions, retains evidence and generates post-flight reports. AI decision support retrieves similar historical flights and their incident evidence; a configured OpenAI or Gemini service may summarize that evidence. AI advice never replaces mandatory validation or Manager decisions.

Scope exclusions: no low-level navigation, stabilization, autonomous flight execution or replacement of GCS/autopilot software. Custom model training and autonomous AI approval are outside scope. Restriction checks cover configured data only; they do not certify compliance with unprovided restrictions.

Flow	Purpose	Primary owner	Features
MF1	Mission Planning & Pre-flight Validation	NghiaHQ	FE-02/03/04
MF2	Real-Time Monitoring & Incident Management	ThongLQ	FE-05/06/07/08/10
MF3	Post-Flight Analysis & Flight History	DucDDA	FE-08/10
MF4	AI-Assisted Mission Risk Assessment	KhanhGNG	FE-09/10

1.1 Product Scope

The system covers mission planning and approval, pre-flight validation, realtime monitoring, incident management, post-flight reporting and AI-assisted decision support. Operations are organized into four mainflows with shared UAV and operator management.

1.2 System Context

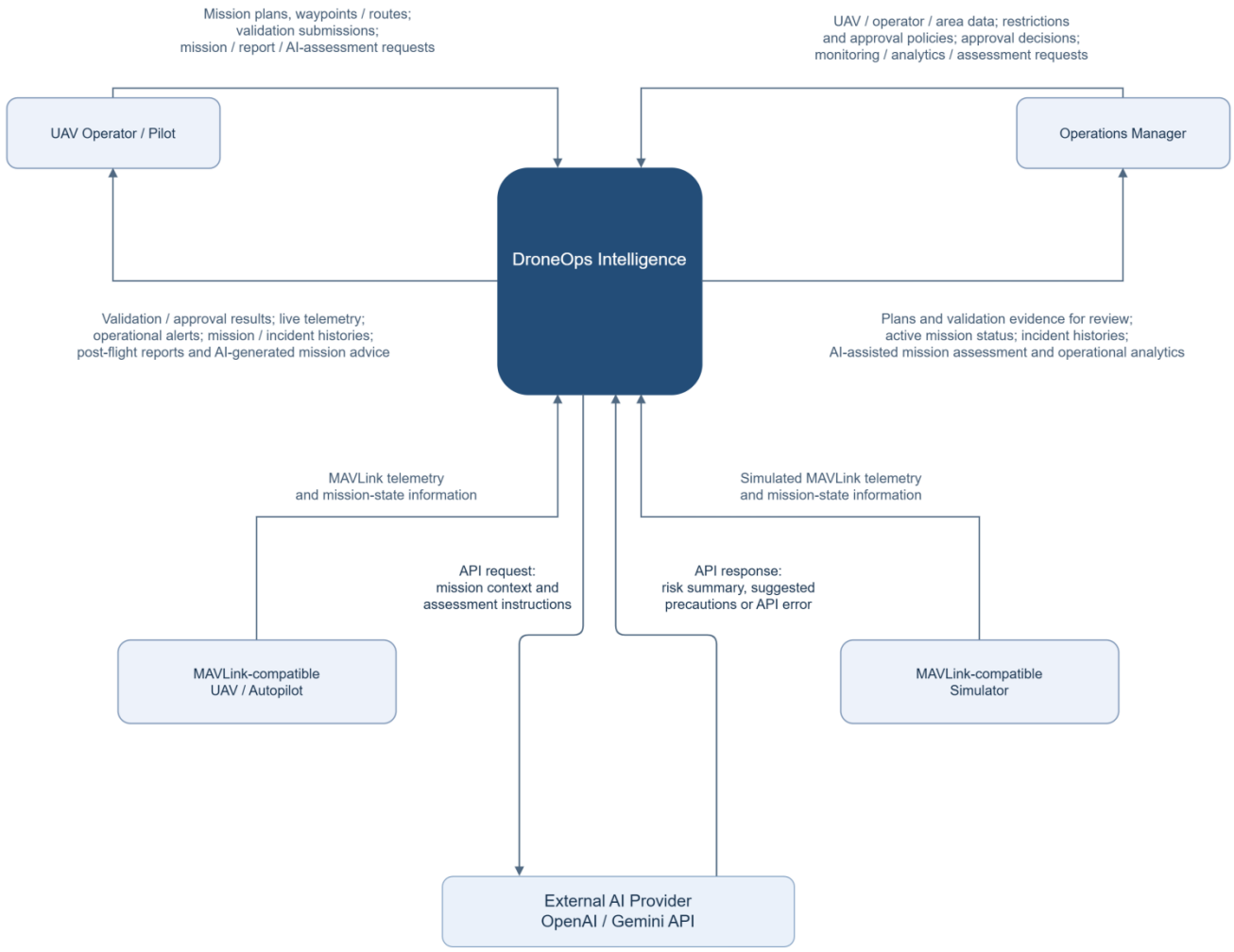


Figure 1. System context diagram.

2. User Requirements

2.1 Actors

ID	Actor	Responsibilities
ACT-01	UAV Operator / Pilot	Create and submit own/assigned missions; review validation, decisions, live telemetry, alerts, history and AI evidence; manage own profile and credentials. Operator Web/mobile scope follows IF-02.
ACT-02	Operations Manager	Manage UAV/operator/area data and restrictions; review and approve/reject missions; monitor operations and review reports/AI advice.
ACT-03	UAV/Autopilot or Simulator	Supply MAVLink telemetry and configured mission-state events. These are alternative compatible data sources; no human login is required.
ACT-04	External AI Provider	Return advisory content or API errors for authorized backend requests to the configured OpenAI or Gemini service.
ACT-05	Pilot Applicant	Register a pilot account, verify the proposed email and provide license evidence; operational access is granted only after Manager approval.

2.2 Use Cases

2.2.1 Diagram(s)

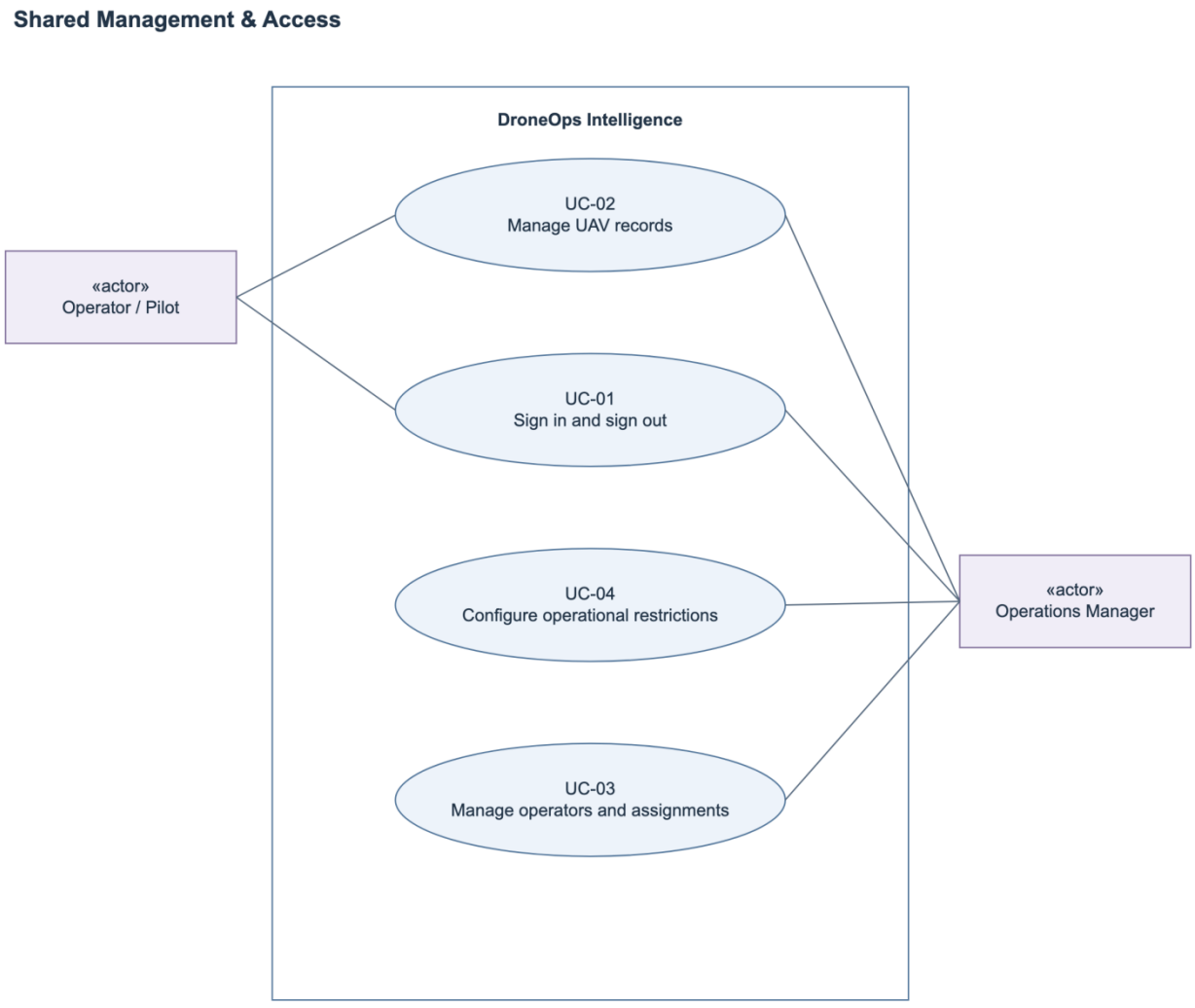


Figure 2. Shared Management & Access

MF1 — Mission Planning & Pre-flight Validation

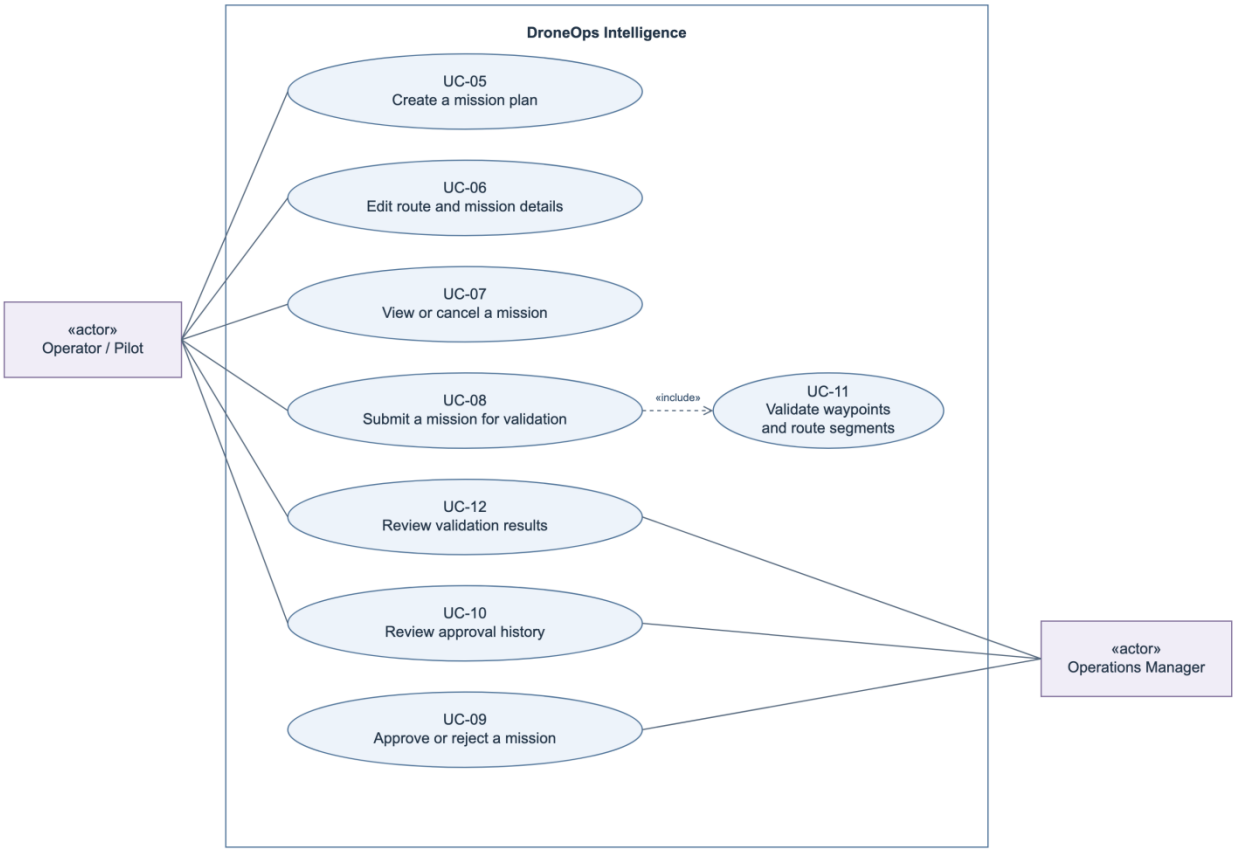


Figure 3. MF1 - Mission Planning & Pre-flight Validation

MF2 - Real-Time Monitoring & Incident Management

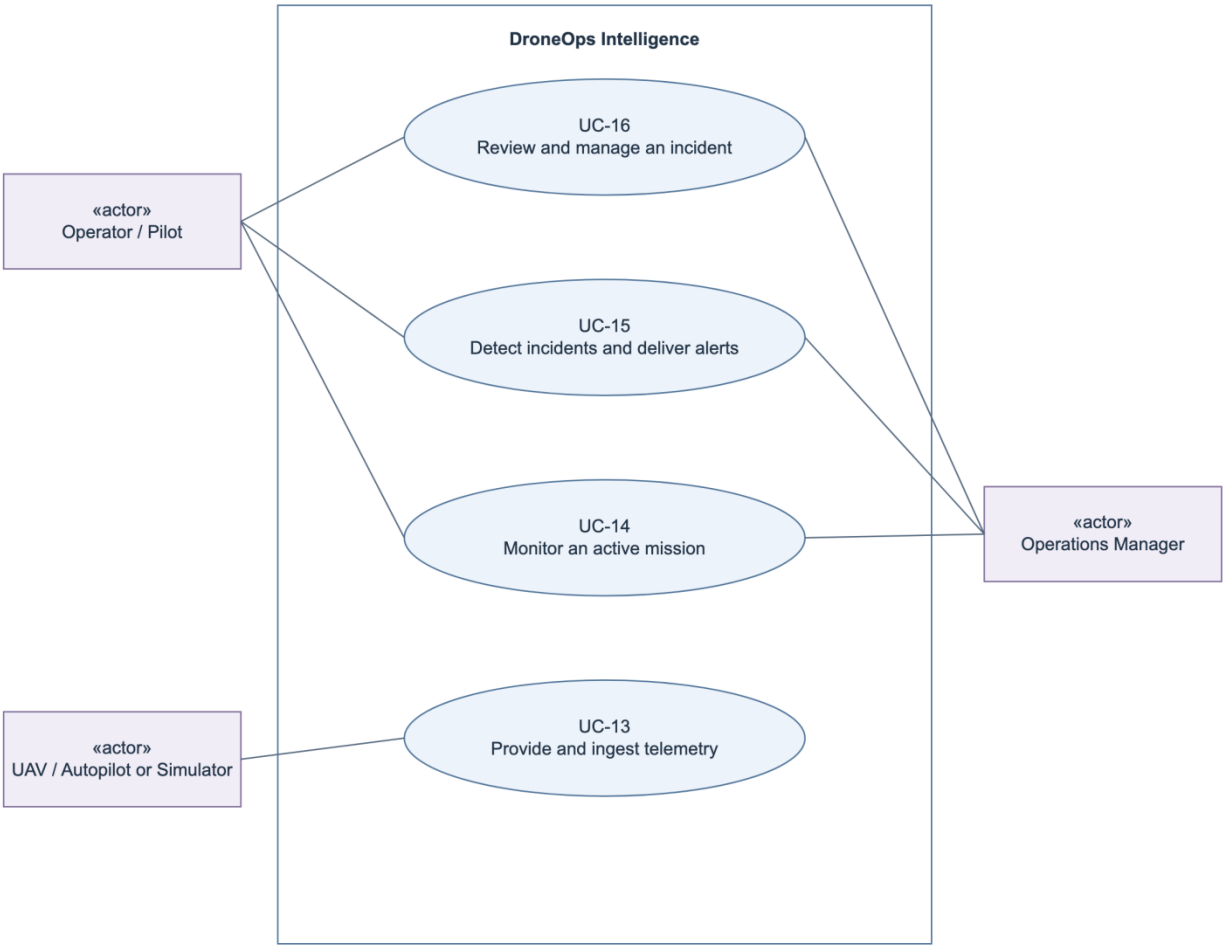


Figure 4. MF2 - Real-Time Monitoring & Incident Management

MF3 - Post-Flight Analysis & Mission History

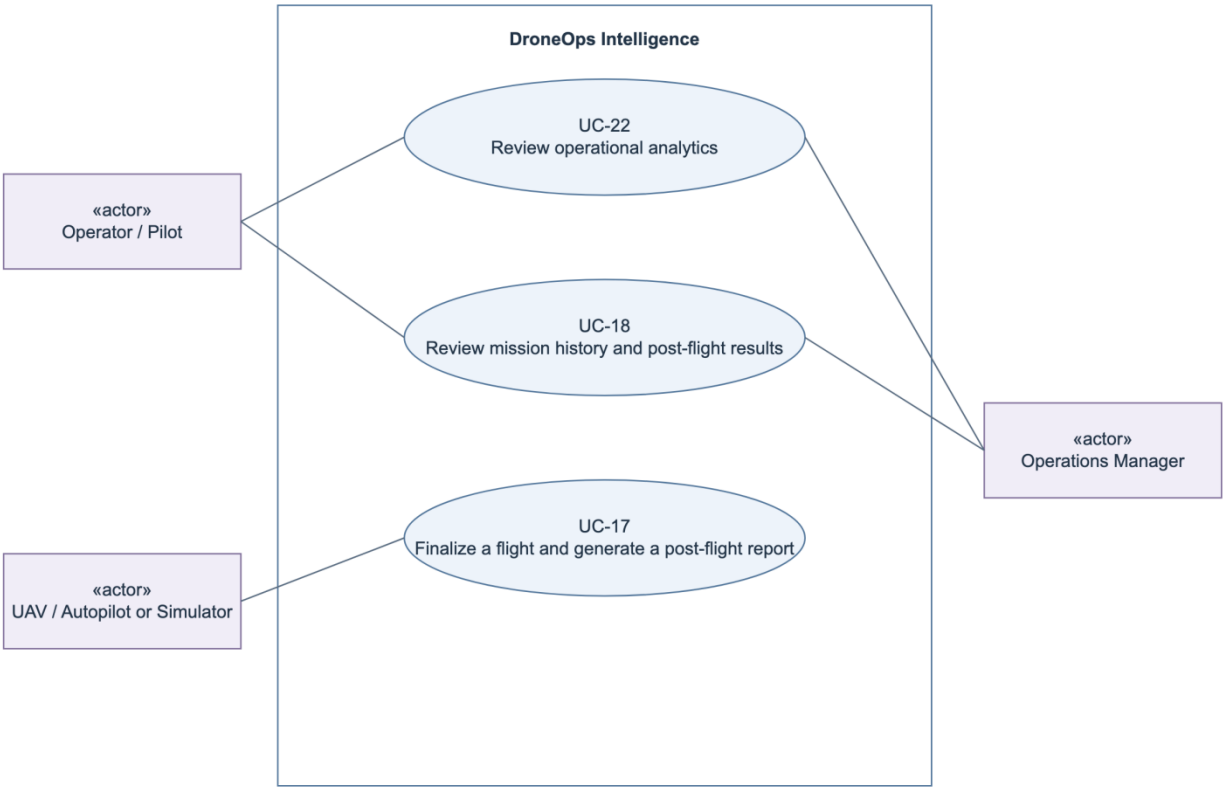


Figure 5. MF3 - Post-Flight Analysis & Flight History

MF4 - AI-Assisted Risk Assessment & Dashboard

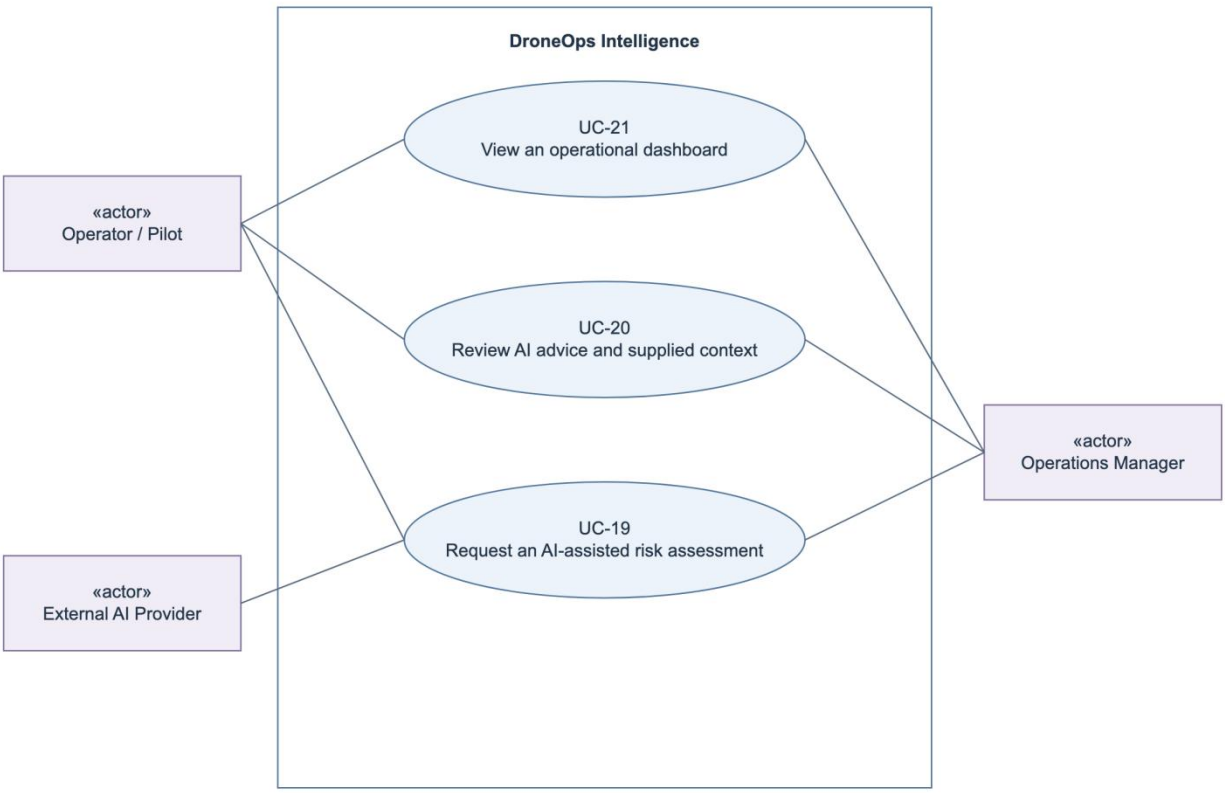


Figure 6. MF4 - AI-Assisted Risk Assessment & Dashboard

2.2.2 Descriptions

ID	Use Case	Actors	Outcome
UC-01	Sign in and sign out	Operator; Manager	Authenticated role-scoped session or signed-out state.
UC-02	Manage UAV records	Manager; Operator (view only)	Saved UAV record and recorded actor/time of change.
UC-03	Manage operators and assignments	Manager	Updated operator/assignment records.
UC-04	Configure operational restrictions	Manager	Versioned operational restrictions available to validation.
UC-05	Create a mission plan	Operator	Draft mission owned by the creator.
UC-06	Edit a route and request mission updates	Operator; Manager	A proposed update stored with author/time and sent to Manager review; current mission content is preserved until approval.
UC-07	View or cancel a mission plan	Operator; Manager	Mission details shown or planning record cancelled with audit evidence.
UC-08	Submit a mission for validation	Operator	Mission/update request visible for Manager review with validation outcome; failed, incomplete or stale checks block approval.
UC-09	Approve or reject a mission	Manager	Immutable decision linked to mission and update request; approval applies proposed content, rejection preserves current content and any existing approval.
UC-10	Review approval history	Operator; Manager	Traceable approval history.
UC-11	Validate waypoints and route segments	Operator (via submission)	Mission/update-request-linked validation result and map evidence.
UC-12	Review mapped validation results	Operator; Manager	User can locate violations and decide whether to edit/resubmit.
UC-13	Provide and ingest telemetry	UAV/Autopilot or Simulator	Stored telemetry and updated UAV/mission projection without flight-control commands.
UC-14	Monitor an active mission	Operator; Manager	Real-time operational view with explicit freshness.
UC-15	Detect incidents and deliver alerts	UAV/Simulator (data source); Operator; Manager	Persisted incident and available in-app alert; no autonomous control action.
UC-16	Review and manage an incident	Operator; Manager	Reviewed incident with append-only status history.
UC-17	Finalize a flight and generate a post-flight report	UAV/Autopilot or Simulator (completion source)	Completed flight history and available or explicitly failed report.
UC-18	Review mission history and post-flight results	Operator; Manager	User reviews recorded flight evidence and report.
UC-19	Request an AI-assisted risk assessment	Operator; Manager	Available assessment or explicit failed status; no change to mission approval.
UC-20	Review AI advice and supplied context	Operator; Manager	Human-reviewed advisory evidence, with manual mission review still available.
UC-21	View an operational dashboard	Operator; Manager	Role-scoped operational overview.
UC-22	Review operational analytics	Operator; Manager	Explainable aggregate view backed by recorded data.

ID	Use Case	Actors	Outcome
UC-23	Register a pilot account and review registration	Applicant; Manager	Verified registration and immutable review outcome; only approved pilots receive operational access.
UC-24	Edit profile and avatar	Operator; Manager (own profile)	Saved profile fields and avatar with visible success/error feedback.
UC-25	Change email with verification	Operator; Manager (own account)	Verified new account email or unchanged current email on any failure.
UC-26	Request and review a license change	Operator; Manager reviewer	Approved replacement or unchanged current credentials with traceable pending/rejected outcome.

3. Functional Requirements

3.1 System Functional Overview

Feature IDs FE-01 to FE-10 preserve Report 1 traceability. UC IDs identify user/system interactions. The screen flow is the source of truth for user-visible navigation: shared entry and account screens lead to role-scoped Operations Manager or UAV Operator lanes. Each detailed function defines normal behavior, validation/exception behavior and its persisted or displayed outcome; background processing is specified separately.

3.1.1 Screen Flow

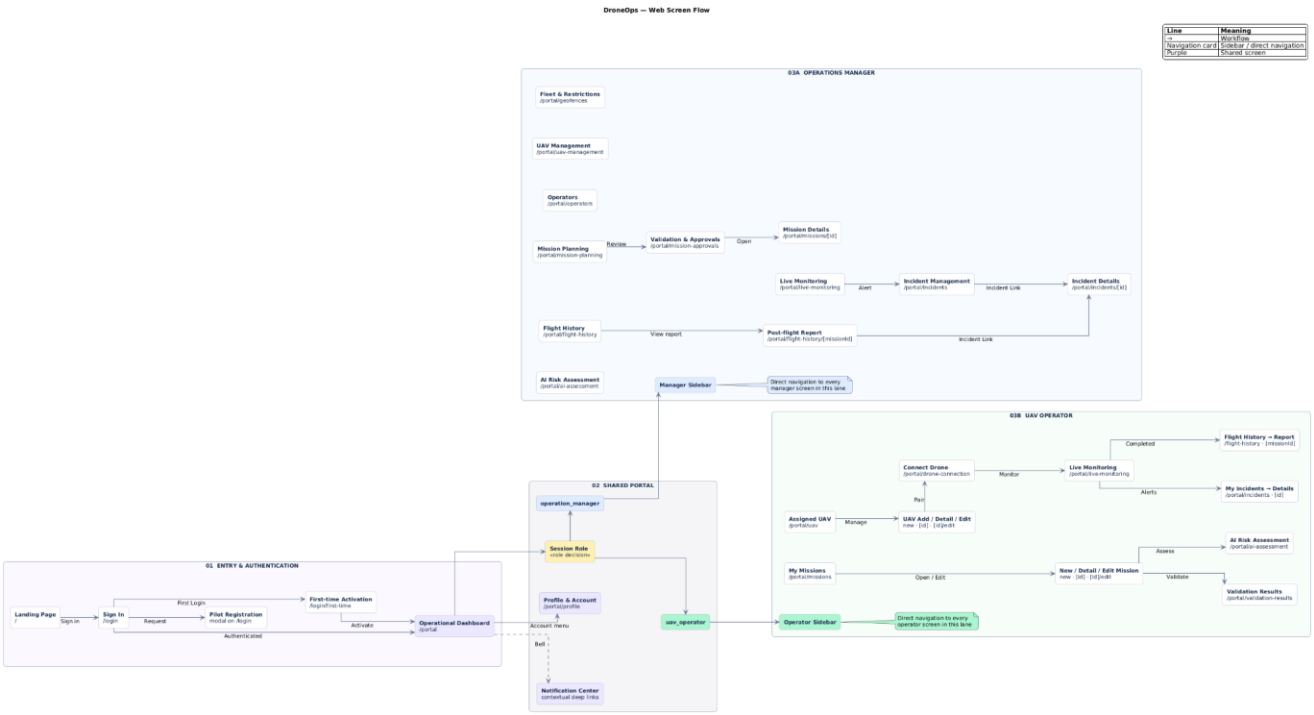


Figure 7. Web screen flow.

3.1.2 Screen Descriptions

ID	Feature	Screen	Route	Description / Navigation
SCR-01	Shared	Landing Page	/	Public entry point; navigates to Sign In.
SCR-02	FE-01	Sign In	/login	Authenticates a user; opens Pilot Registration on request and First-time Activation on first login.
SCR-03	FE-01	Pilot Registration	Modal on /login	Collects a pilot application and verification data before account review.
SCR-04	FE-01	First-time Activation	/login/first-time	Completes first-login activation, then opens the Operational Dashboard.
SCR-05	FE-10	Operational Dashboard	/portal	Authenticated landing screen; resolves the session role and exposes role-scoped navigation.
SCR-06	Shared	Profile & Account	/portal/profile	Allows either role to view and maintain its own account and profile.
SCR-07	Shared	Notification Center	Contextual deep links	Opened from the dashboard bell; links the user to an authorized related record.

SCR-08	FE-01	Operators	/portal/operators	Manager screen for operator review, status and assignments.
SCR-09	FE-01	UAV Management	/portal/uav-management	Manager screen for fleet-wide UAV records and status.
SCR-10	FE-01/04	Fleet & Restrictions	/portal/geofences	Manager screen for fleet context, geofences and operational restrictions.
SCR-11	FE-02	Mission Planning	/portal/mission-planning	Manager planning workspace; sends a mission to Validation & Approvals for review.
SCR-12	FE-03/04	Validation & Approvals	/portal/mission-approvals	Manager queue for validation evidence and approval decisions; opens Mission Details.
SCR-13	FE-02/03/04	Mission Details	/portal/missions/[id]	Manager view of mission content, validation and decision history.
SCR-14	FE-05/06/07	Live Monitoring	/portal/live-monitoring	Role-scoped live mission view; links alerts to incidents and completion to history.
SCR-15	FE-07/08	Incident Management	/portal/incidents	Manager incident list and operational management screen.
SCR-16	FE-08	Incident Details	/portal/incidents/[id]	Detailed incident evidence and permitted lifecycle actions.
SCR-17	FE-08	Flight History	/portal/flight-history	Manager list of completed flights; opens a Post-flight Report.
SCR-18	FE-08	Post-flight Report	/portal/flight-history/[missionId]	Manager report view with links to related incident details.
SCR-19	FE-09	AI Risk Assessment	/portal/ai-assessment	Role-scoped advisory risk assessment for an authorized mission.
SCR-20	FE-01	Assigned UAV	/portal/uav	Operator view of assigned UAVs; opens UAV create/detail/edit where permitted.
SCR-21	FE-01	UAV Add / Detail / Edit	/portal/uav/new · /portal/uav/[id] · /portal/uav/[id]/edit	Operator UAV workflow; pairs the selected UAV through Connect Drone.
SCR-22	FE-05	Connect Drone	/portal/drone-connection	Pairs/connects an authorized UAV and proceeds to Live Monitoring.
SCR-23	FE-02	My Missions	/portal/missions	Operator mission list; opens mission create/detail/edit.
SCR-24	FE-02	New / Detail / Edit Mission	/portal/missions/new · /portal/missions/[id] · /portal/missions/[id]/edit	Operator mission workflow; launches validation or AI assessment.
SCR-25	FE-04	Validation Results	/portal/validation-results	Shows the result and evidence produced by validating a mission.
SCR-26	FE-07/08	My Incidents / Details	/portal/incidents · /portal/incidents/[id]	Operator-scoped incident list and details reached from live alerts.
SCR-27	FE-08	Flight History / Report	/flight-history · /flight-history/[missionId]	Operator-scoped completed-flight history and report.

3.1.3 Screen Authorization

Authorization policy: Operators access missions they own or are assigned to; Managers access records within their managed operational scope. Both UI and backend enforce these rules. Administrative registration/license review is assigned to the Operations Manager role; an implementation role named admin receives these actions only through explicit authorization. Read access does not grant permission to modify evidence.

Screen / Action	UAV Operator	Operations Manager
Landing Page; Sign In; Pilot Registration; First-time Activation	Public for entry/registration; activation requires a valid first-login challenge.	Same authentication entry; Manager privilege is never self-selected.
Operational Dashboard; Profile & Account; Notification Center	Own account and own/assigned deep links only.	Own account and managed-scope deep links only.
Operators	No administrative access; own profile is handled in Profile & Account.	Manage operator review, status and assignments within scope.
UAV Management	No fleet-wide management access.	Manage UAV records and status within scope.
Fleet & Restrictions	View relevant restrictions through authorized mission context; no edits.	Create and revise geofences/restrictions; changes trigger stale-validation policy.
Mission Planning; Validation & Approvals; Manager Mission Details	No access to Manager-only screens.	Plan/review managed missions; approval requires fresh passing validation and separation of duties.
Assigned UAV; UAV Add / Detail / Edit; Connect Drone	Access assigned UAVs and permitted own UAV workflow; connect only an authorized configured UAV.	No Operator-lane navigation; fleet administration uses UAV Management.
My Missions; New / Detail / Edit Mission; Validation Results	Create and maintain own/assigned pre-flight missions and review their validation results.	Use Manager mission screens for managed records and decisions.
Live Monitoring	View own/assigned active mission and follow alerts/completion links.	View active missions within managed scope.
Incident Management / My Incidents; Incident Details	View and acknowledge own/assigned incidents; cannot resolve.	View, acknowledge and resolve managed incidents with required reason/note.
Flight History; Post-flight Report	View own/assigned completed flights and reports.	View managed flights; retry/regenerate reports with audit history.
AI Risk Assessment	Generate/retry/view for own/assigned missions; cannot overwrite provider output.	Generate/retry/view for managed missions; cannot overwrite provider output.

Managers may request AI assessments, change dashboard filters and propose mission edits within their scope. Neither role may rewrite measured telemetry, historical flight evidence or the original provider response. The author of a mission update cannot approve that same update.

3.1.4 Non-Screen Functions

ID	Feature	Function	Description
NSF-01	FE-04	Route validation	UC-11; evaluate waypoints and connecting segments, persist result/evidence.
NSF-02	FE-05	Telemetry gateway/normalization	UC-13; receive supported MAVLink, validate identity/data, normalize and persist.
NSF-03	FE-05/06	State projection and live delivery	UC-13/14; update latest valid state and publish to authorized clients.
NSF-04	FE-07	Rule evaluation and connectivity timer	UC-15; evaluate packet-driven and time-driven abnormal conditions.
NSF-05	FE-07/08	Incident persistence and alert delivery	UC-15/16; preserve evidence and deliver in-app notifications.
NSF-06	FE-08	Flight finalization and report generation	UC-17; recognize configured completion and publish history/report status.
NSF-07	FE-09	External AI API adapter	UC-19; capture context, request advice, validate response and handle failures.
NSF-08	FE-10	Analytics aggregation	UC-22; aggregate authorized recorded history with documented formulas.
NSF-09	Shared	Authorization and material-change audit	Validate identity/action/record scope and record actor/time for changes.
NSF-10	FE-01	Verification and private media	Deliver protected email challenges, rate-limit verification, authorize avatar/license storage access and preserve current data on failure.
NSF-11	FE-09	Historical similarity retrieval	Build normalized feature representations, retrieve scoped Top-k completed-flight evidence and retain retrieval provenance.

3.1.5 Entity Relationship Diagram

The entity relationship diagrams describe the mission planning and operational data model. A mission stores the current plan; a separate update request stores proposed changes pending Manager review. Validation, decisions, telemetry, incidents and reports retain their relationship to the relevant mission.

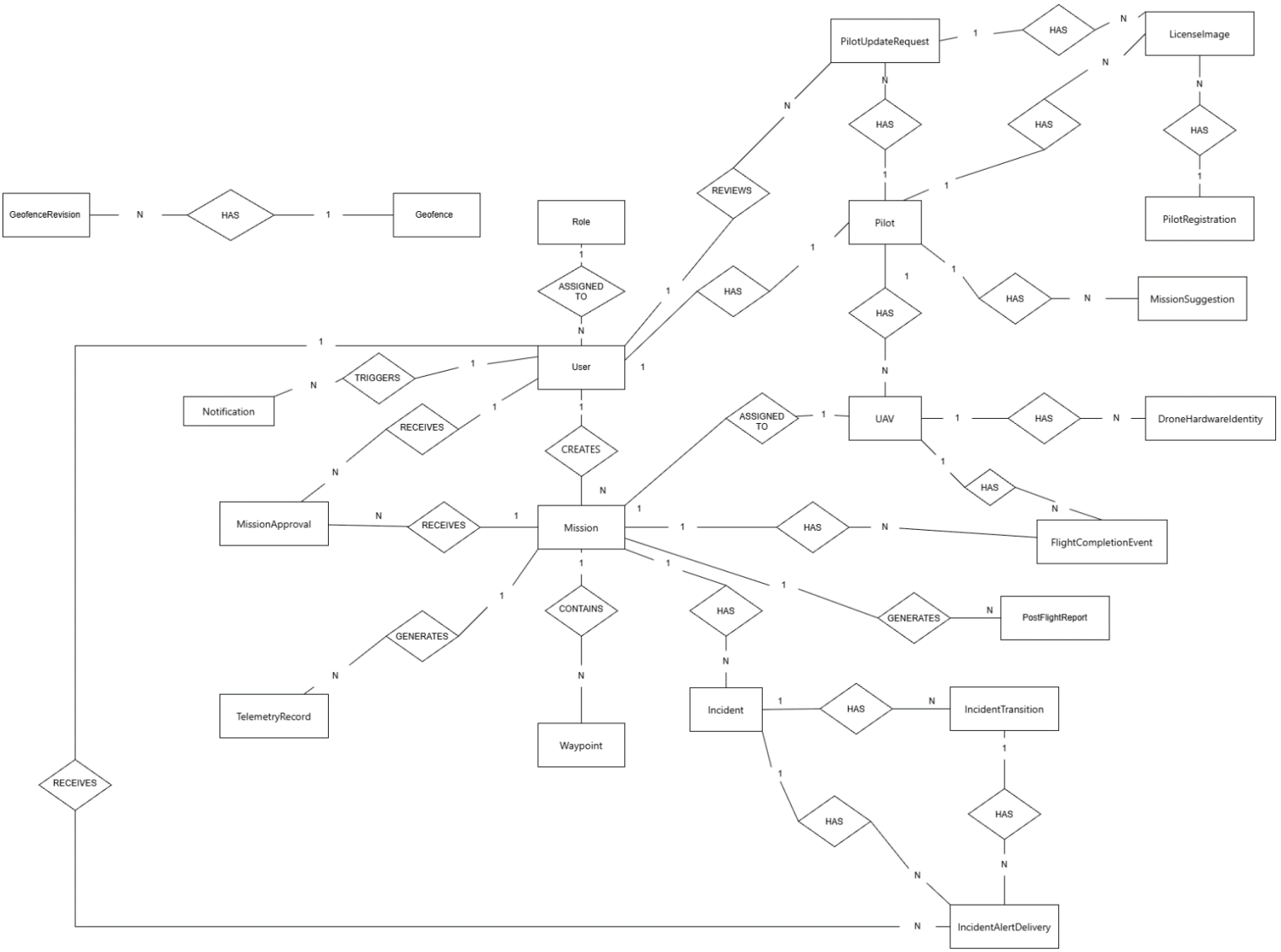


Figure 8. Entity relationship diagram.

No.	Entity	Domain	Description
1	Role	Identity and access	Defines an authorization category assigned to user accounts and used to control access to system capabilities.
2	User	Identity and access	Represents an authenticated system account and the actor behind mission, approval, review, and notification activities.
3	Pilot	Identity and access	Stores the operational pilot profile associated with a user and links that profile to aircraft, evidence, requests, and suggestions.
4	PilotRegistration	Identity and access	Represents a pilot onboarding or registration submission together with its supporting license evidence.
5	PilotUpdateRequest	Identity and access	Captures a controlled request to update pilot information and supports review and attachment of evidence before changes are accepted.
6	LicenseImage	Identity and access	Stores or references license-document imagery used as evidence for a pilot profile, registration, or profile update request.
7	UAV	Fleet and airspace	Represents an aircraft in the managed fleet, including its pilot assignment, mission assignment, hardware identities, and completion events.
8	DroneHardwareIdentity	Fleet and airspace	Represents a hardware-level identity associated with a UAV, such as a serial, device, controller, or communication identity.

No.	Entity	Domain	Description
9	Geofence	Fleet and airspace	Represents a logical operational or restricted area whose definition can evolve through versioned revisions.
10	GeofenceRevision	Fleet and airspace	Represents one historical or current revision of a geofence, preserving changes to geometry or operational rules over time.
11	Mission	Mission lifecycle	Acts as the central operational record for a planned flight and connects its creator, aircraft, route, approvals, telemetry, incidents, completion events, and reports.
12	Waypoint	Mission lifecycle	Represents an ordered route point contained by a mission and used to define the intended flight path.
13	MissionApproval	Mission lifecycle	Records an approval review received by a mission and associated with the user responsible for the review activity.
14	MissionSuggestion	Mission lifecycle	Represents a mission-related recommendation or planning suggestion associated with a pilot.
15	TelemetryRecord	Mission lifecycle	Represents telemetry generated during a mission, allowing operational and post-flight analysis of aircraft state.
16	FlightCompletionEvent	Mission lifecycle	Records a completion-related lifecycle event connected to both the mission and the UAV that performed the flight.
17	PostFlightReport	Mission lifecycle	Represents a report generated for a mission after flight activity, consolidating outcome and supporting evidence.
18	Incident	Incidents and notifications	Represents an operational anomaly or safety event linked to a mission and its subsequent lifecycle and alert deliveries.
19	IncidentTransition	Incidents and notifications	Records a change in an incident's lifecycle state so that status history and operational actions remain traceable.
20	IncidentAlertDelivery	Incidents and notifications	Tracks delivery of an incident alert to a user, with context from the incident and its transition history.
21	Notification	Incidents and notifications	Represents a user-facing system notification. The source diagram labels its relationship to User as TRIGGERS and should be reviewed for intended direction.

3.1.6 Mainflow Activity Diagrams

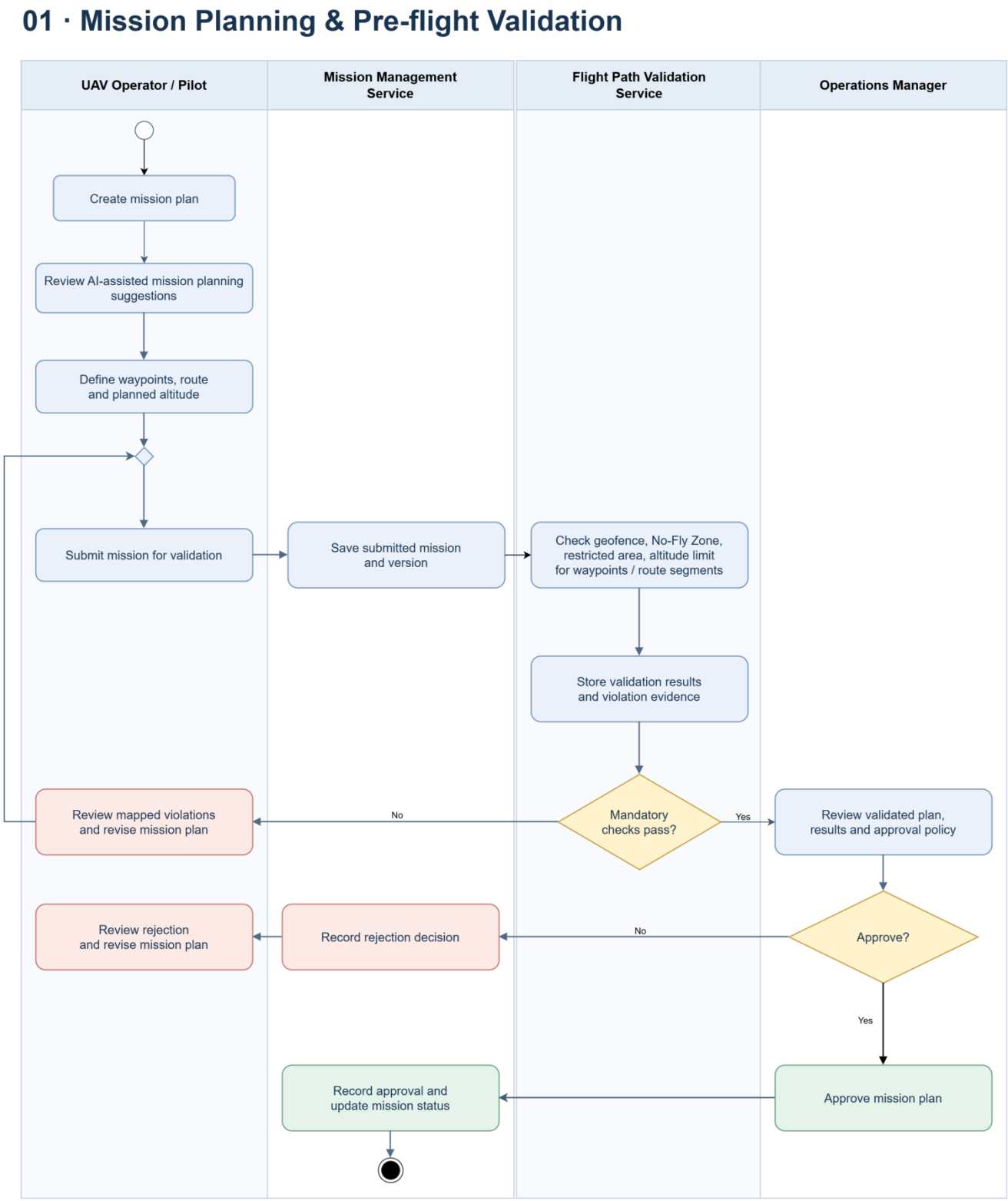


Figure 9. Mission Planning & Pre-flight Validation.

02 · AI-Assisted Mission Risk Assessment

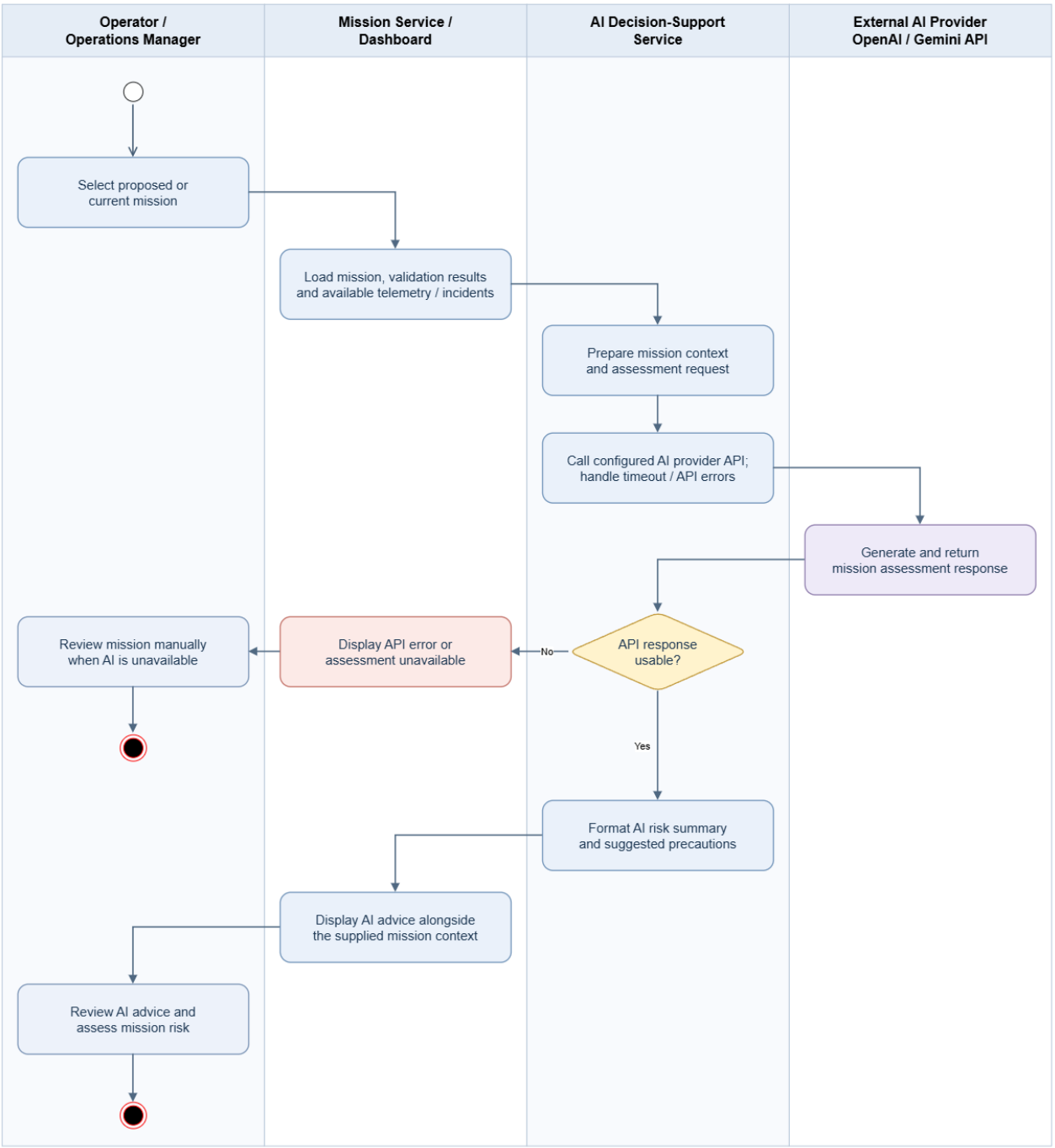


Figure 10. AI-Assisted Mission Risk Assessment.

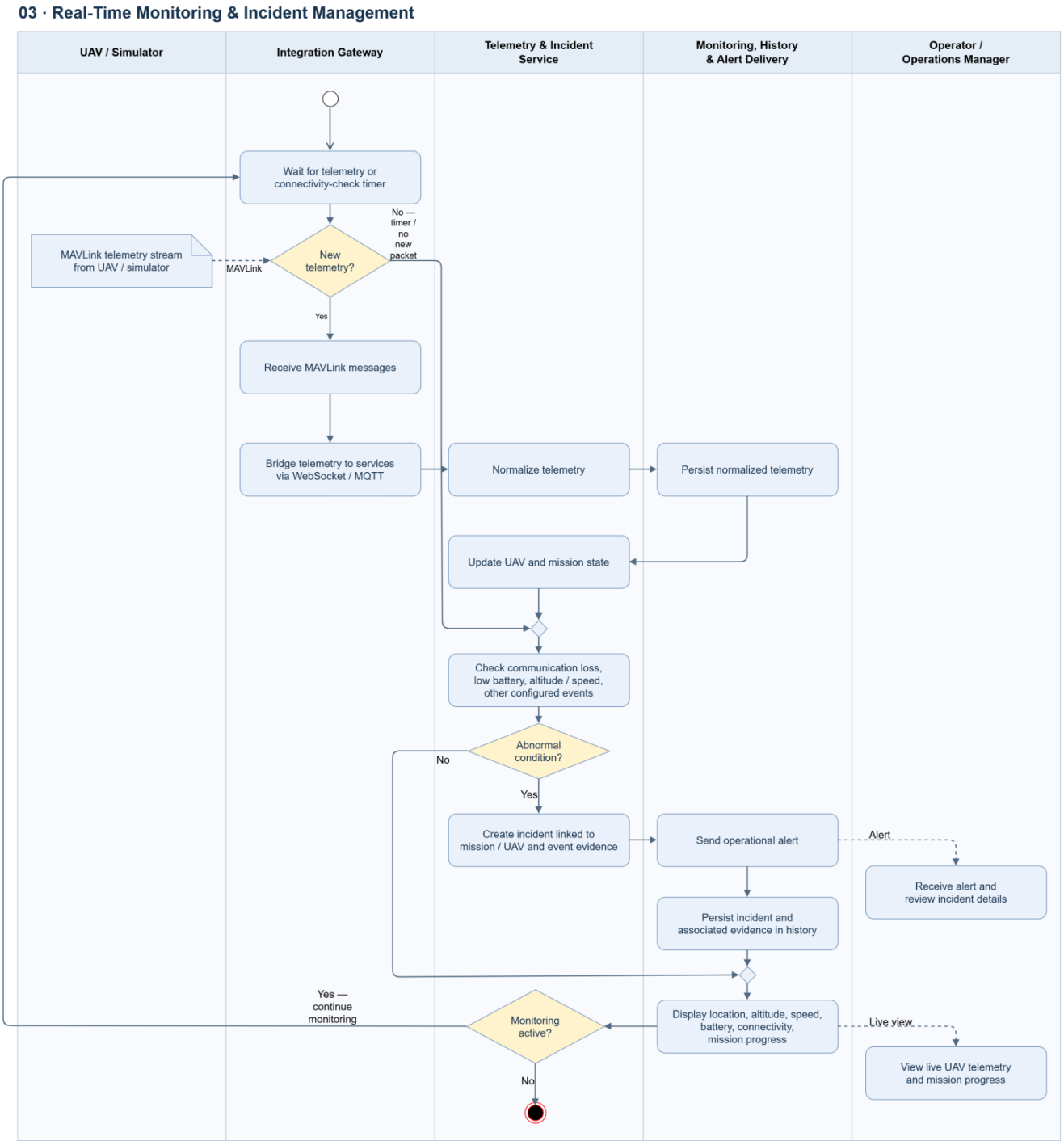


Figure 11. Real-Time Monitoring & Incident Management.

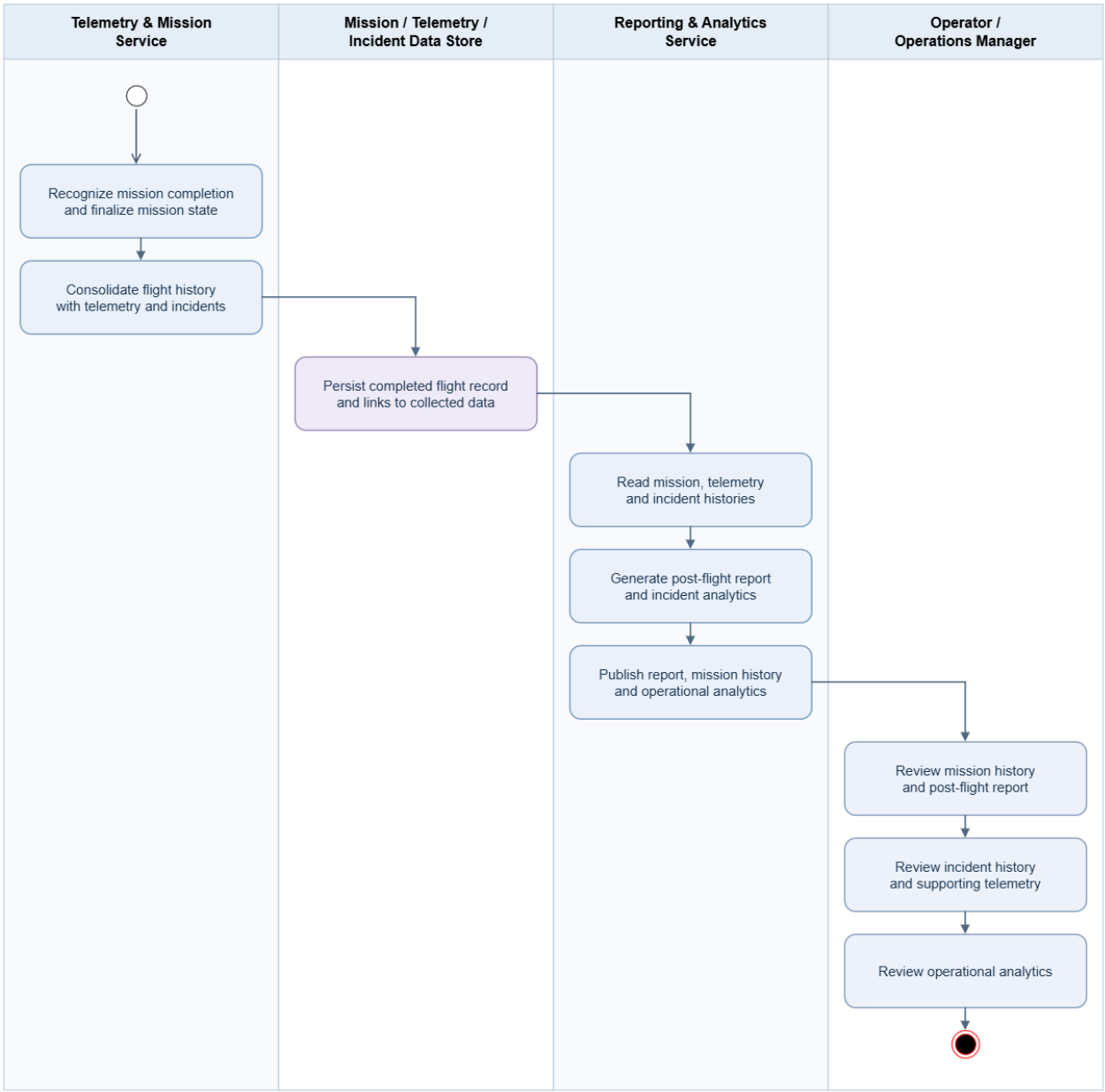


Figure 12. Post-Flight Analysis & Flight History

3.1.7 Lifecycle Diagrams

A new mission begins in Draft. Initial submission triggers validation and Manager review. Every subsequent accepted pre-flight edit creates a pending update for the same mission and automatically triggers validation and review. The current mission remains unchanged while review is pending. The lifecycle and concurrency rules are defined in BR-13 and CR-05.

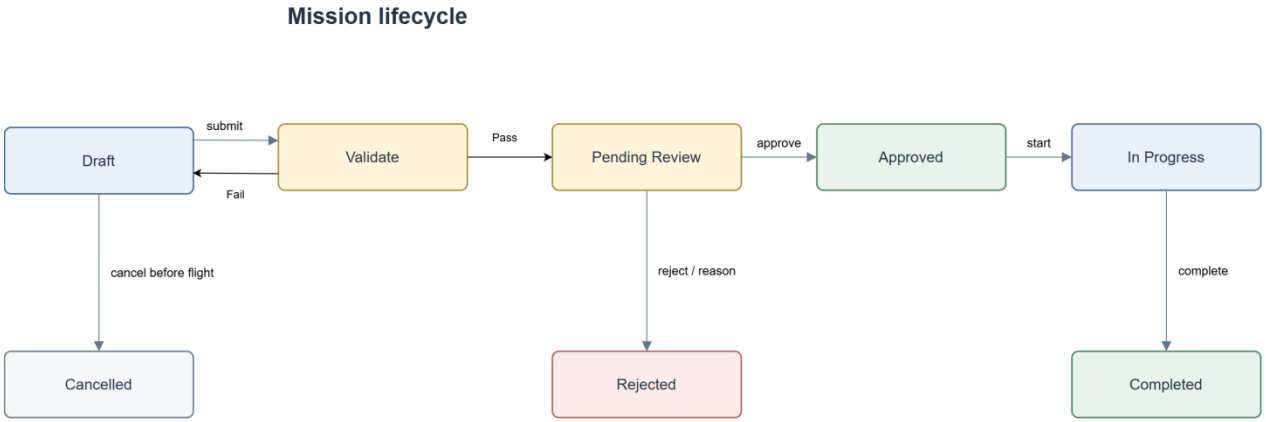


Figure 13. Mission lifecycle.

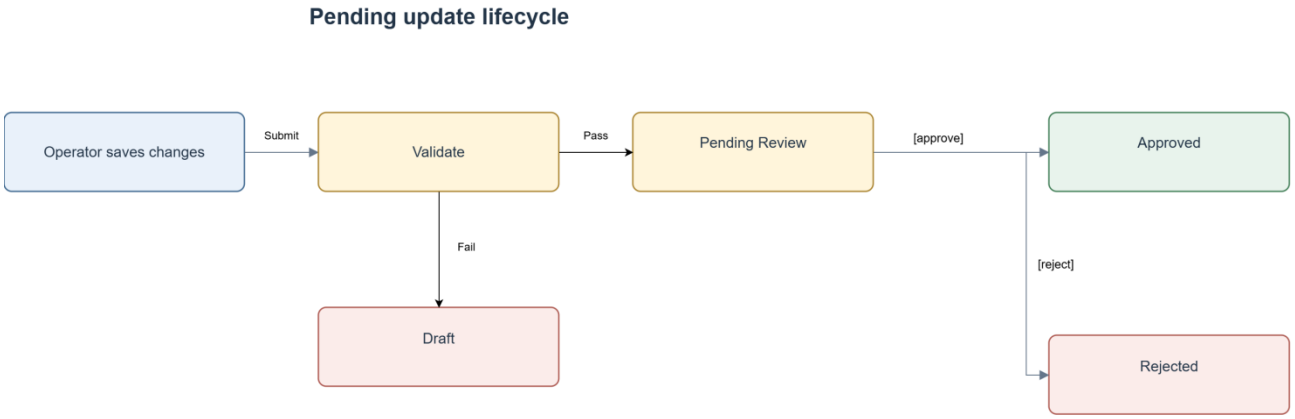


Figure 14. Pending update request lifecycle.

Approval atomically applies proposed content to the same mission ID. Rejecting an update preserves current content and operational status, including an existing Approved status. A mission with a pending update cannot begin execution. Cancellation affects the planning record and never sends a flight-control command.

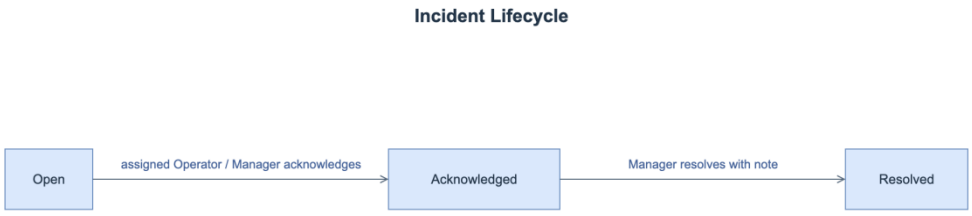


Figure 15. Incident lifecycle.

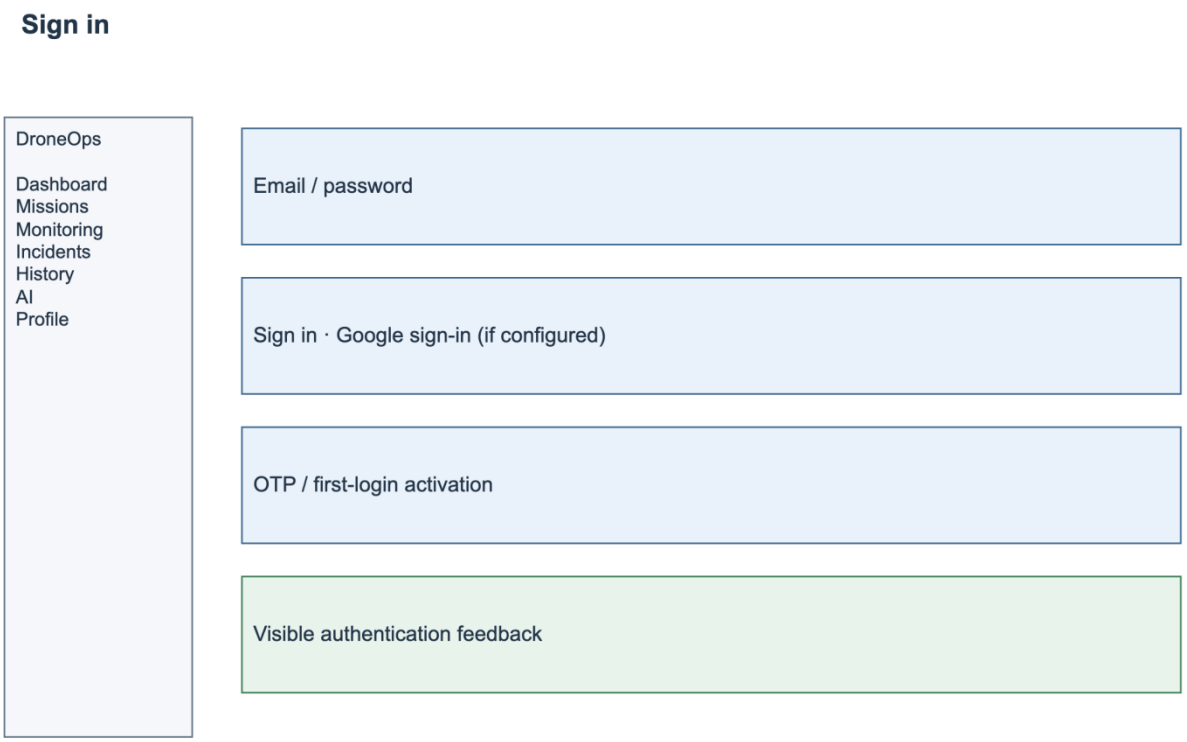
3.2 FE-01 — UAV and Operator Management

3.2.1 Sign in and sign out (UC-01)

Specification	Details
Actors / Interface	Operator; Manager | SCR-02
Trigger	Open the application or select Sign out.
Preconditions	An account is available for sign-in. Sign-out requires an active session.

Specification	Details
Input / Data	Account identifier and password.
FR-01.1 — Normal flow	The system shall support the following behavior: Verify credentials and active account status; create a session; show only permitted navigation. Sign out invalidates the session.
FR-01.2 — Validation / Alternative flow	The system shall enforce the following checks and outcomes: Invalid credentials show MSG-02 without identifying which field is wrong; inactive account or expired session is denied; unauthorized direct URL/API access is rejected.
FR-01.3 — Postcondition	The system shall provide: Authenticated role-scoped session or signed-out state.
Business rules	BR-01; CR-01
Traceability	UC-01

Screen layout — SCR-02



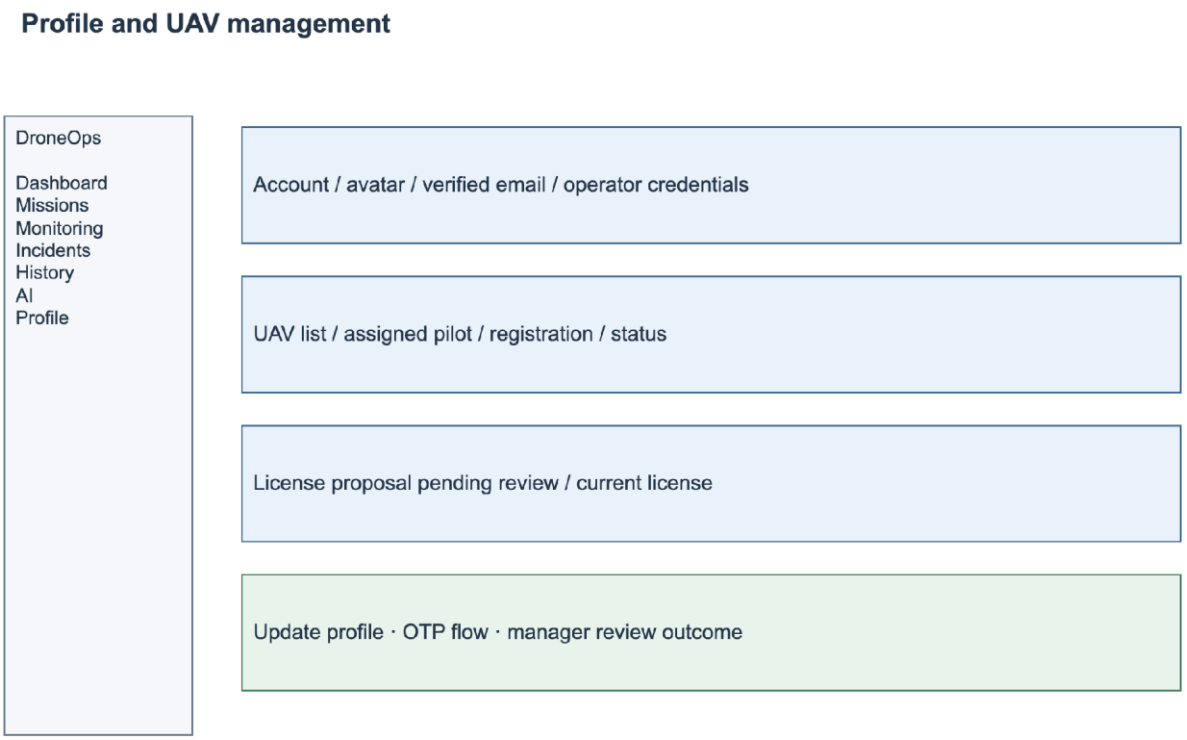
SCR-01 — Screen layout.

3.2.2 Manage UAV records (UC-02)

Specification	Details
Actors / Interface	Manager; Operator (view only) | SCR-09 / SCR-20 / SCR-21
Trigger	UAV Management (/portal/uav-management) > Add / Edit / Deactivate.
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Unique UAV code, name, model, operational status and configured telemetry identity.
FR-02.1 — Normal flow	The system shall support the following behavior: List/search UAVs; validate required/unique identifiers; save a UAV record; keep historical mission references when deactivating. Operators can view assigned UAV information.

Specification	Details
FR-02.2 — Validation / Alternative flow	Duplicate identities or invalid data shall be rejected. Conflicting updates require a reload. An operator or UAV assigned to an active mission cannot be deactivated.
FR-02.3 — Postcondition	The system shall provide: Saved UAV record and recorded actor/time of change.
Business rules	BR-02; BR-12
Traceability	UC-02

Screen layout — SCR-09 / SCR-20 / SCR-21



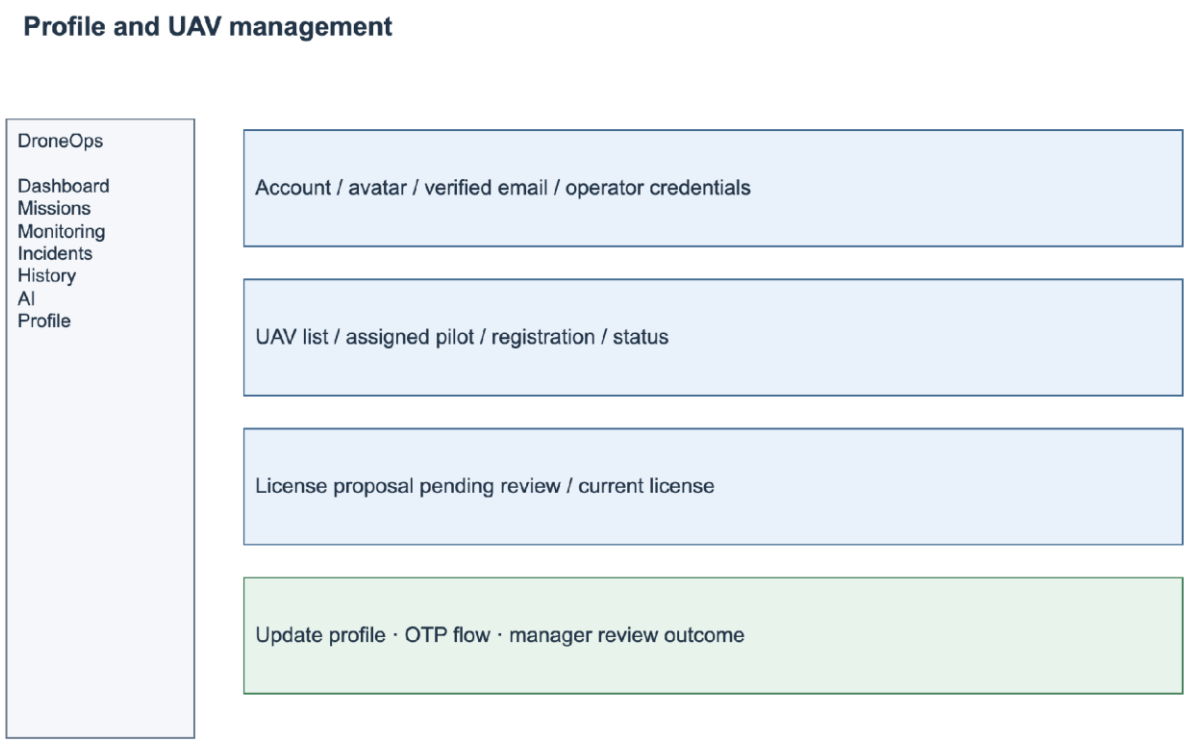
SCR-03 — Screen layout.

3.2.3 Manage operators and assignments (UC-03)

Specification	Details
Actors / Interface	Manager | SCR-08
Trigger	Operators (/portal/operators) > Add / Edit / Assign.
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Operator name, account identifier, status and mission/UAV assignment.
FR-03.1 — Normal flow	The Manager shall review pilot registrations, activate or disable eligible accounts, maintain operator assignments and retain past assignment history. Operators manage their own profile through UC-24. Only explicitly authorized administrators may provision Manager privileges.

Specification	Details
FR-03.2 — Validation / Alternative flow	Duplicate account identifiers and inactive or ineligible assignees shall be rejected. The last active Manager cannot be disabled or demoted. A pilot with an expired, revoked or unapproved license cannot be assigned to a new flight or start a mission. Conflicting UAV/operator schedules follow BR-14.
FR-03.3 — Postcondition	The system shall provide: Updated operator/assignment records.
Business rules	BR-01; BR-02
Traceability	UC-03

Screen layout — SCR-08



SCR-04 — Screen layout.

3.2.4 Configure operational restrictions (UC-04)

Specification	Details
Actors / Interface	Manager | SCR-10
Trigger	Fleet & Restrictions (/portal/geofences).
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Area name, polygon/coordinates, restriction type, altitude limit and applicable reference/units.
FR-04.1 — Normal flow	The system shall support the following behavior: Create/view/edit configured operational areas, geofences, No-Fly Zones and restricted areas; validate geometry; save a restriction revision for subsequent validation.

Specification	Details
FR-04.2 — Validation / Alternative flow	Reject empty or self-intersecting polygons, fewer than three distinct vertices, invalid coordinates, invalid altitude intervals and missing altitude reference. Persist each restriction revision. A revision affecting a route invalidates prior validation and prevents flight start until fresh validation and Manager approval are recorded. In-progress flights retain their execution snapshot and display the updated restriction warning.
FR-04.3 — Postcondition	The system shall provide: Versioned operational restrictions available to validation.
Business rules	BR-03; BR-04
Traceability	UC-04

Screen layout — SCR-10



SCR-05 — Screen layout.

3.2.5 Register a pilot account and review registration (UC-23)

Specification	Details
Actors / Interface	Applicant; Manager | SCR-03 / SCR-08
Trigger	Open Register; Manager opens Registration Requests.
Preconditions	Registration is public; review requires an authenticated authorized Manager.
Input / Data	Full name, unique email, password, phone, date of birth, brief flight experience, license number/expiry and JPEG/PNG/WebP license image up to 5 MB.
FR-23.1 — Normal flow	Validate fields, verify email using a one-time code and create Pending Review registration. Manager reviews identity and license evidence, approves or rejects with a reason, and records the decision. Approval activates Operator access; pending/rejected accounts cannot access operational functions.

Specification	Details
FR-23.2 — Validation / Alternative flow	Reject duplicate normalized email, invalid phone/date, expired license or unsupported upload. Email verification uses a six-digit code valid for 10 minutes, at most five attempts and a 60-second resend interval; resending invalidates the prior code. A rejected application may be corrected and resubmitted without creating a second account. Review retries are idempotent; applicants cannot select Manager privileges.
FR-23.3 — Postcondition	Verified registration and immutable review outcome; only approved pilots receive operational access.
Business rules	BR-01; BR-12; BR-16; BR-17; CR-01; CR-02; CR-05
Traceability	UC-23

3.2.6 Edit profile and avatar (UC-24)

Specification	Details
Actors / Interface	Operator; Manager (own profile) | SCR-06
Trigger	Profile > Edit details or Change avatar.
Preconditions	Authenticated account.
Input / Data	Phone, date of birth, brief flight experience and optional avatar image.
FR-24.1 — Normal flow	Allow the user to update their own phone, date of birth and brief flight experience, and replace their avatar. Show the saved avatar in the profile and authorized Manager operator roster/detail. Use the label Briefly describe your flight experience.
FR-24.2 — Validation / Alternative flow	Require a nonfuture birth date and valid phone; do not accept profile fields that change role, account status or approved license. Accept JPEG/PNG/WebP up to 5 MB, validate file content and retain the previous avatar on failure. Stale updates require reload.
FR-24.3 — Postcondition	Saved profile fields and avatar with visible success/error feedback.
Business rules	BR-01; BR-12; BR-16; BR-17; CR-01; CR-02; CR-05
Traceability	UC-24

3.2.7 Change email with verification (UC-25)

Specification	Details
Actors / Interface	Operator; Manager (own account) | SCR-06
Trigger	Profile > Change email > Send verification code > Confirm.
Preconditions	Authenticated account; proposed email differs from the current one.
Input / Data	New email and verification code.
FR-25.1 — Normal flow	Reject an already registered normalized email before sending a code. Send the code to the new address and visibly state the destination. Keep the current email until confirmation succeeds. On success atomically reserve the unique new email, update the account and invalidate pending codes.
FR-25.2 — Validation / Alternative flow	Use a six-digit code valid for 10 minutes, maximum five verification attempts and a 60-second resend interval. A resend invalidates the previous code. Scope the challenge to account and proposed email; store a protected verifier, never the raw code. Delivery failure leaves the current email unchanged. Recheck uniqueness on confirmation and require reauthentication after change.
FR-25.3 — Postcondition	Verified new account email or unchanged current email on any failure.
Business rules	BR-01; BR-12; BR-16; BR-17; CR-01; CR-02; CR-05

Specification	Details
Traceability	UC-25

3.2.8 Request and review a license change (UC-26)

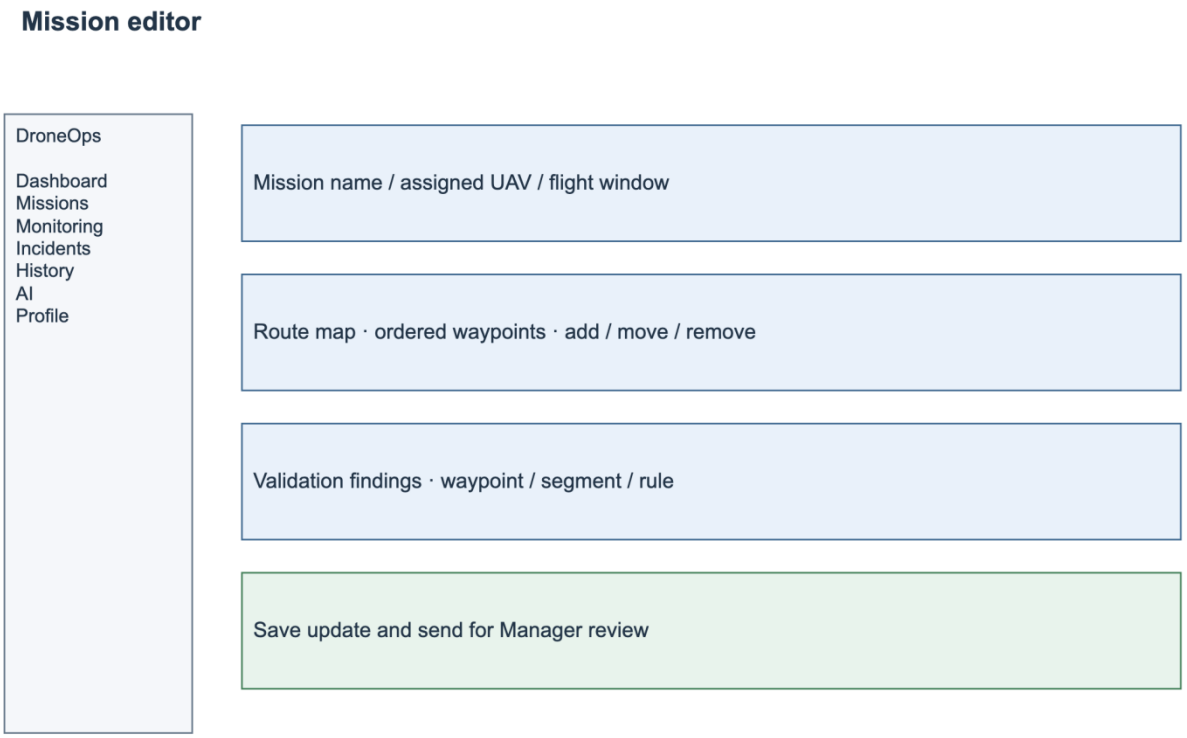
Specification	Details
Actors / Interface	Operator; Manager reviewer | SCR-06 / SCR-08
Trigger	Profile > Flight credentials > Request change; Manager > License Changes.
Preconditions	Authenticated pilot for submission; scoped Manager other than request author for decision.
Input / Data	Proposed license number, expiry date and private supporting image.
FR-26.1 — Normal flow	Store proposed credentials and image separately and create Pending Review request. Display current license and request status to the pilot. Manager sees the pilot avatar/profile, current and proposed credentials and accessible images; Approve atomically replaces current credentials, Reject retains them and requires a reason. Notify the pilot in-app and retain decision history.
FR-26.2 — Validation / Alternative flow	Require a future expiry, nonblank license number and valid JPEG/PNG/WebP up to 5 MB. Permit one pending request per account; a replacement supersedes the prior pending request and cannot be approved from a stale screen. Failed upload cannot create an incomplete request. Rejection does not extend the old license validity.
FR-26.3 — Postcondition	Approved replacement or unchanged current credentials with traceable pending/rejected outcome.
Business rules	BR-01; BR-12; BR-16; BR-17; CR-01; CR-02; CR-05
Traceability	UC-26

3.3 FE-02 — Mission Planning and Management

3.3.1 Create a mission plan (UC-05)

Specification	Details
Actors / Interface	Operator | SCR-23 / SCR-24
Trigger	Missions > New Mission.
Preconditions	The Operator is authenticated and an eligible assigned UAV is available.
Input / Data	Name, purpose, UAV/operator, operational area and proposed schedule if used.
FR-05.1 — Normal flow	Create a Draft mission with one mission ID and the selected owner, UAV, operational area and scheduled start/end. The creation form may save an initial ordered route without review. After creation, accepted changes follow UC-06; initial submission follows UC-08. No mission versions are created.
FR-05.2 — Validation / Alternative flow	Require a name, active UAV, eligible assigned pilot, valid operational area and an end time later than the start time. Reject conflicting UAV/pilot bookings. A route requires at least two ordered waypoints before submission. Warn before navigating away from unsaved form changes.
FR-05.3 — Postcondition	The system shall provide: Draft mission owned by the creator.
Business rules	BR-02; BR-05
Traceability	UC-05

Screen layout — SCR-23 / SCR-24



SCR-07 — Screen layout.

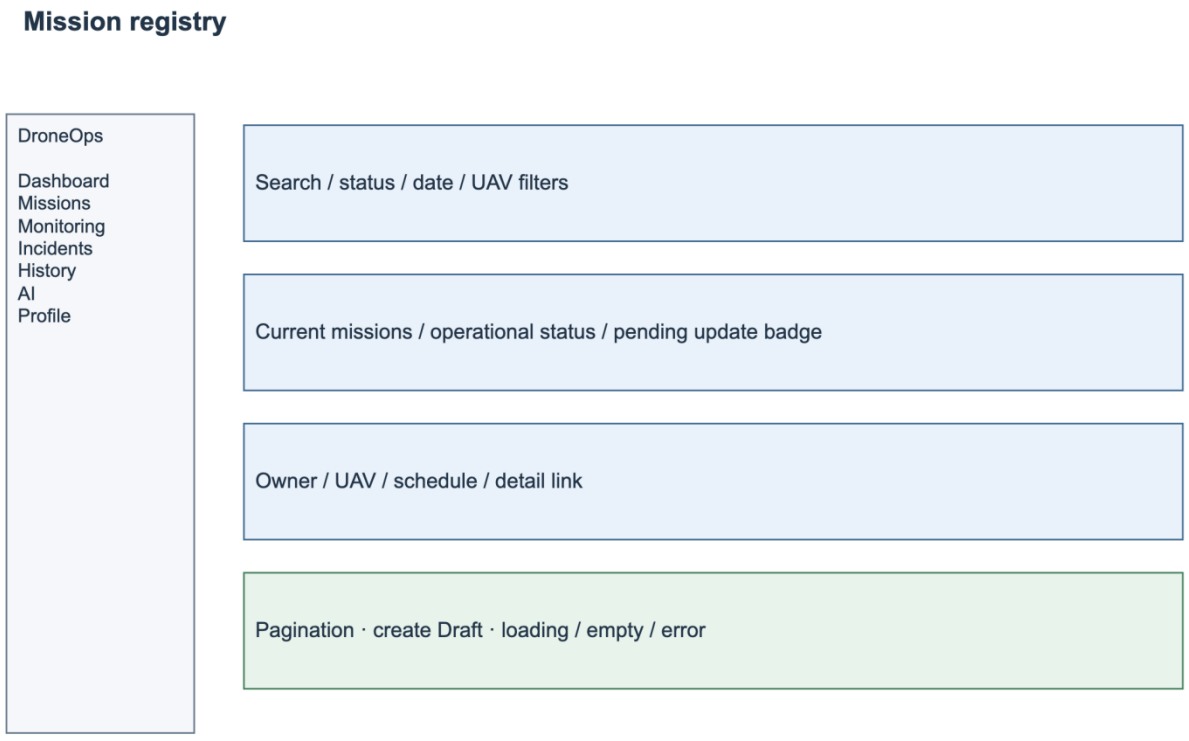
3.3.2 Edit a route and request mission updates (UC-06)

Specification	Details
Actors / Interface	Operator; Manager | SCR-24 / SCR-13
Trigger	Missions > Mission Details > Edit Plan.
Preconditions	The mission exists within the user's scope and has not entered execution.
Input / Data	Ordered waypoint coordinates and planned altitude; mission metadata.
FR-06.1 — Normal flow	An authorized Operator or Manager may propose metadata, schedule, assignment and route changes to an eligible pre-flight mission. Each accepted Save stores a complete proposed payload separately, creates a request identity, runs validation and automatically sends the request to Manager review. The current mission content and status remain unchanged until approval; no second manual Submit is required.
FR-06.2 — Validation / Alternative flow	Reject changes to In Progress, Completed or Cancelled missions, malformed coordinates, invalid assignments and schedule conflicts. Only one pending update may exist per mission. A later accepted Save atomically supersedes the earlier pending request, preserves its history and invalidates its evidence; the superseded request cannot be decided. A stale form or decision returns a conflict and requires reload. Proposed content cannot reuse validation for different content.
FR-06.3 — Postcondition	The system shall provide: Proposed update stored with author/time and automatically sent to Manager; current mission remains unchanged until approval.
Business rules	BR-04; BR-05; BR-06
Traceability	UC-06

3.3.3 View or cancel a mission plan (UC-07)

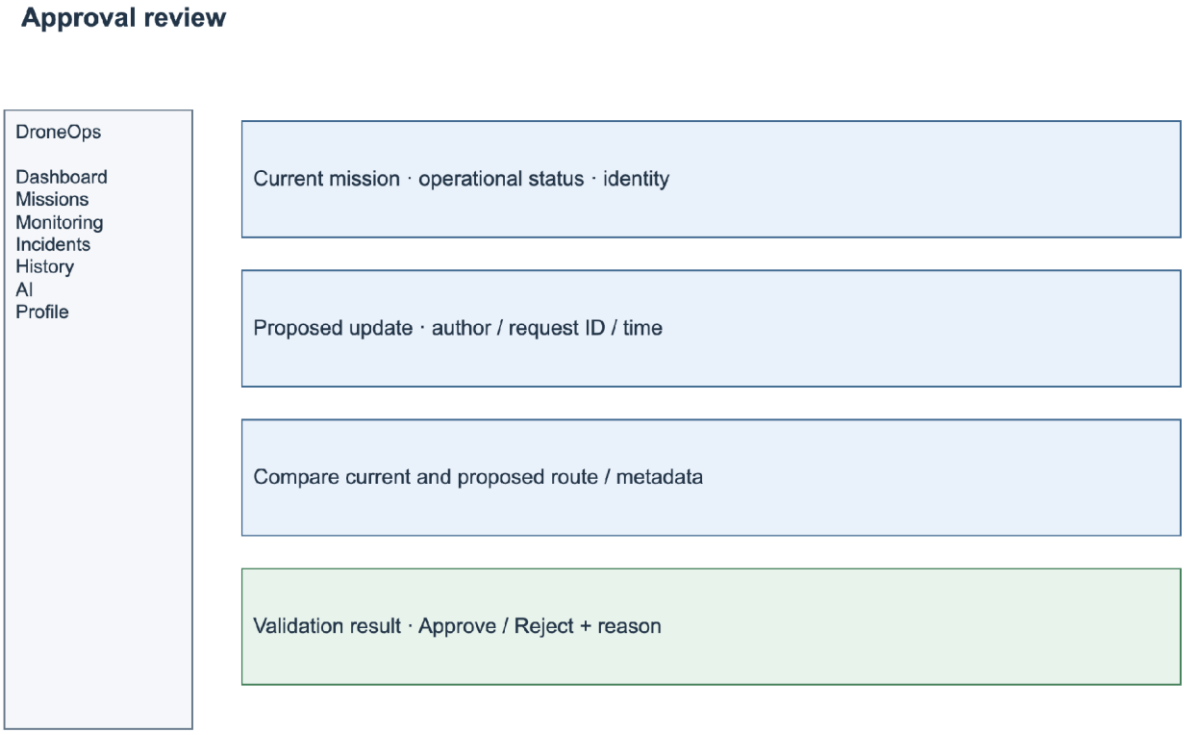
Specification	Details
Actors / Interface	Operator; Manager | SCR-23 / SCR-24 / SCR-13
Trigger	Missions > select mission; choose Cancel if permitted.
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Mission ID, search/filter and cancellation reason.
FR-07.1 — Normal flow	The system shall support the following behavior: Display current mission, proposed updates, validation and decisions. If cancellation is permitted, confirm the action, preserve history and mark the plan Cancelled.
FR-07.2 — Validation / Alternative flow	Draft, Pending Review, Rejected and Approved missions may be cancelled by an authorized owner/assigned Operator or scoped Manager with a nonblank reason. Cancellation closes any pending submission/update as Cancelled, retains decisions and releases future bookings. In Progress, Completed and already Cancelled missions cannot be cancelled by this action.
FR-07.3 — Postcondition	The system shall provide: Mission details shown or planning record cancelled with audit evidence.
Business rules	BR-01; BR-05; BR-06
Traceability	UC-07

Screen layout — SCR-23 / SCR-24 / SCR-13



SCR-06 — Screen layout.

Screen layout — SCR-23 / SCR-24 / SCR-13



SCR-08 — Screen layout.

3.4 FE-03 — Mission Approval Workflow

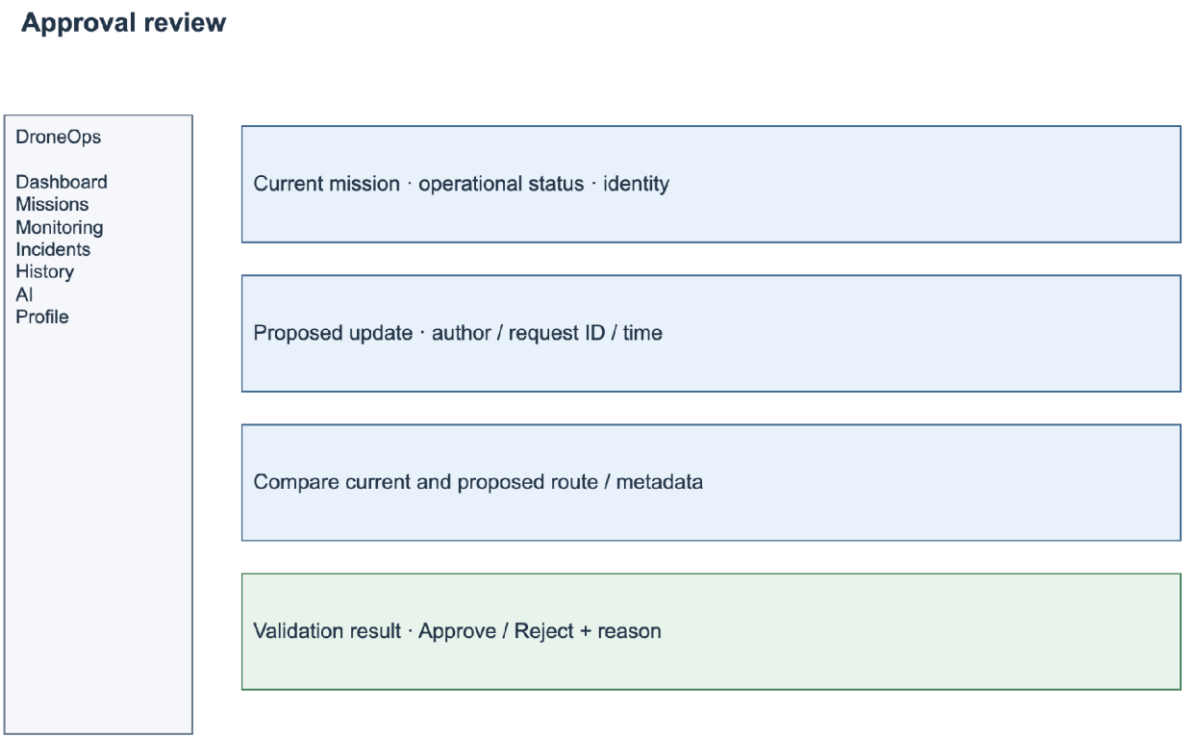
3.4.1 Submit a mission for validation (UC-08)

Specification	Details
Actors / Interface	Operator | SCR-24 / SCR-25
Trigger	Mission Details > Submit.
Preconditions	The mission has a saved route and required assignment data.
Input / Data	Current mission or proposed update request with route and required assignment data.
FR-08.1 — Normal flow	For an initial Draft or Rejected mission, Submit validates required fields, assignments and schedule, captures the submitted plan, invokes UC-11 and moves the mission to Pending Review. For an edit to an existing mission, Save already creates the review request and invokes validation through UC-06. Failed, incomplete or stale results remain visible to the Manager and block approval; they do not authorize flight.
FR-08.2 — Validation / Alternative flow	Missing required fields prevent acceptance. Validation service failure produces Error or Incomplete, never Pass. Retrying the same operation identity returns the existing result rather than creating another submission. Initial submitted content cannot be overwritten in place; a subsequent edit follows the reviewed-update policy and supersedes the previous pending review.
FR-08.3 — Postcondition	The system shall provide: Mission/update request visible to Manager with validation outcome; failed/incomplete/stale checks block approval.
Business rules	BR-03; BR-05; BR-06
Traceability	UC-08

3.4.2 Approve or reject a mission (UC-09)

Specification	Details
Actors / Interface	Manager | SCR-12 / SCR-13
Trigger	Validation & Approvals > select pending mission.
Preconditions	The Manager is authenticated and the submission or update request is awaiting review.
Input / Data	Current mission, proposed content, update request ID, validation evidence and decision reason.
FR-09.1 — Normal flow	The Manager shall compare current and submitted/proposed content and check fresh mandatory validation and assignment eligibility. For an initial submission, Approve changes the mission to Approved and Reject changes it to Rejected. For an update, Approve atomically applies the proposed content to the same mission ID and sets Approved; Reject changes only the request to Rejected and preserves current content and mission status. Record reviewer, time, reason and reviewed payload/evidence identity; notify the Operator in-app.
FR-09.2 — Validation / Alternative flow	Require a nonblank rejection reason. The author of a proposed update cannot approve it; another authorized Manager must decide it. Pending requests may be rejected even when validation failed, but approval requires Pass against current restrictions and unchanged proposed content. Stale, superseded, cancelled or already decided requests cannot receive a conflicting decision. Repeated identical decisions are idempotent. Concurrent content/restriction changes force refresh and revalidation.
FR-09.3 — Postcondition	The system shall provide: Immutable decision linked to mission and review request; approve applies proposed content, reject preserves existing content and prior approval.
Business rules	BR-03; BR-05; BR-06
Traceability	UC-09

Screen layout — SCR-12 / SCR-13



SCR-09 — Screen layout.

3.4.3 Review approval history (UC-10)

Specification	Details
Actors / Interface	Operator; Manager | SCR-13 / SCR-24
Trigger	Mission Details > Updates & Decisions.
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Mission ID and selected update request ID, where applicable.
FR-10.1 — Normal flow	The system shall support the following behavior: Show submission, validation, decision, reviewer/time and rejection reasons in order; scope operator access to permitted missions.
FR-10.2 — Validation / Alternative flow	The system shall enforce the following checks and outcomes: No prior decision displays an empty state; inaccessible mission/update request is denied.
FR-10.3 — Postcondition	The system shall provide: Traceable approval history.
Business rules	BR-01; BR-06
Traceability	UC-10

3.5 FE-04 — Geofence and Flight Path Validation

3.5.1 Validate waypoints and route segments (UC-11)

Specification	Details
Actors / Interface	Operator (via submission) | SCR-25 / background
Trigger	Submission or an authorized revalidation request.
Preconditions	The mission route and active configured restrictions are available for validation.
Input / Data	Current mission route or proposed update payload and current configured restrictions.
FR-11.1 — Normal flow	Validate all waypoints and their connecting straight segments against active operational boundaries, No-Fly Zones, restricted polygons and altitude limits. Use WGS84 latitude/longitude in decimal degrees and metres above mean sea level (MSL) for planned altitude. A route touching or crossing a prohibited polygon boundary is a violation; an operational-area boundary is included. Persist Pass/Fail/Incomplete/Error, affected waypoint/segment, rule, restriction revision and checked payload identity.
FR-11.2 — Validation / Alternative flow	Latitude shall be within [-90, 90] and longitude within [-180, 180]. Require at least two waypoints and unique contiguous sequence numbers. Polygon rings must be closed, non-self-intersecting and contain at least three distinct vertices. Altitude limits are inclusive; no-flight restrictions override allowed areas. AGL-only restrictions require an available terrain/reference conversion to MSL; missing conversion or restriction data produces Incomplete. Empty configured prohibited-zone results are distinct from an unavailable restriction service.
FR-11.3 — Postcondition	The system shall provide: Mission/update-request-linked validation result and map evidence.
Business rules	BR-03; BR-04
Traceability	UC-11

3.5.2 Review mapped validation results (UC-12)

Specification	Details
Actors / Interface	Operator; Manager | SCR-25 / SCR-13
Trigger	Validation Results.
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Validation run ID, mission ID and optional update request ID.
FR-12.1 — Normal flow	The system shall support the following behavior: Highlight failed waypoints/segments on the map; list violated configured rule and explanation; show pass/fail/incomplete and validation timestamp.
FR-12.2 — Validation / Alternative flow	The system shall enforce the following checks and outcomes: Map provider failure leaves a textual violation list accessible; an old run is labelled outdated.
FR-12.3 — Postcondition	The system shall provide: User can locate violations and decide whether to edit/resubmit.
Business rules	BR-03; BR-06
Traceability	UC-12

3.6 FE-05 — MAVLink Telemetry Integration

3.6.1 Provide and ingest telemetry (UC-13)

Specification	Details
Actors / Interface	UAV/Autopilot or Simulator | Non-screen
Trigger	Configured source sends MAVLink data to the integration gateway.
Preconditions	A configured telemetry source identifies the UAV and supplies a timestamped sample.
Input / Data	Source identity, message timestamp and supported location/altitude/speed/battery/state fields.
FR-13.1 — Normal flow	The system shall support the following behavior: Accept a configured source; normalize units/fields; persist timestamps and validity; associate data with the correct UAV and, when known, flight; update latest valid state and publish an update.
FR-13.2 — Validation / Alternative flow	Reject malformed packets and unknown source/UAV mappings; retain diagnostics without treating missing values as zero. Deduplicate using source identity, boot/session identity, message type and source sequence/time. Exact replay creates no second persisted event. Out-of-order samples may be retained as historical evidence but cannot replace newer live state or create retrospective live alerts. A configured UAV may map to at most one active flight; ambiguous data remains unassociated and cannot advance a mission state.
FR-13.3 — Postcondition	The system shall provide: Stored telemetry and updated UAV/mission projection without flight-control commands.
Business rules	BR-07; BR-08
Traceability	UC-13

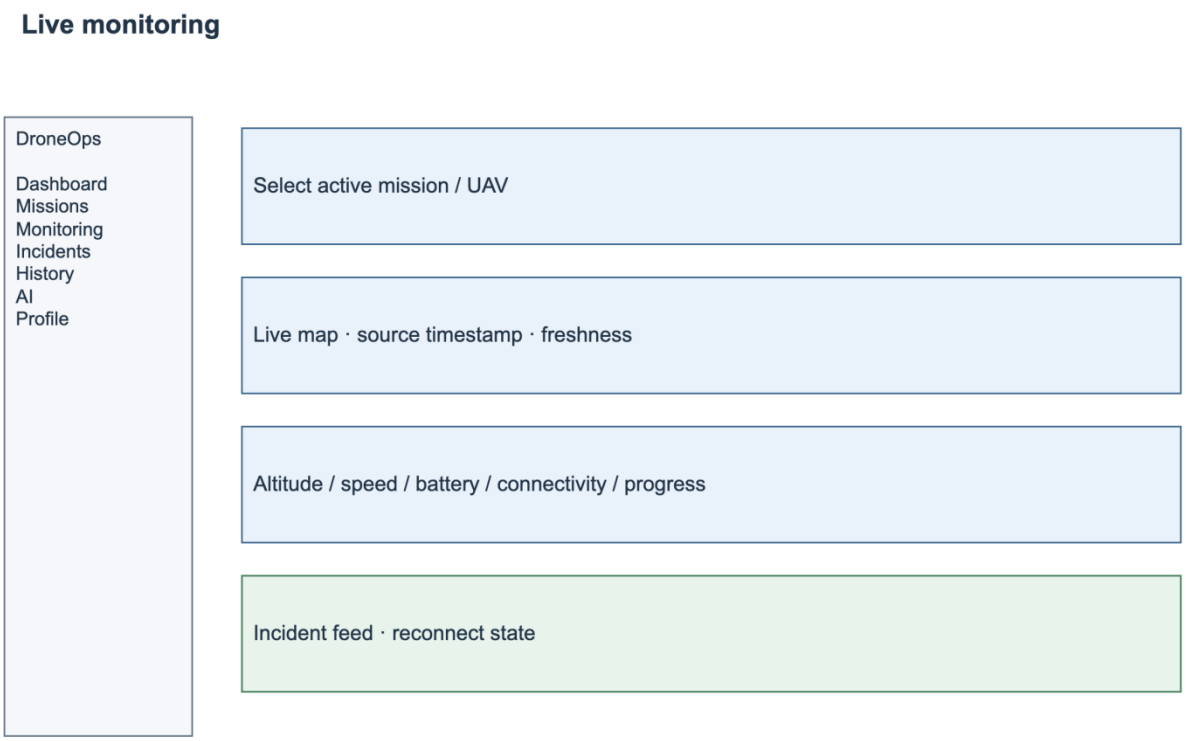
3.7 FE-06 — Real-Time Mission Monitoring

3.7.1 Monitor an active mission (UC-14)

Specification	Details
Actors / Interface	Operator; Manager | SCR-14

Specification	Details
Trigger	Live Monitoring > select active mission/UAV.
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Authorized mission/UAV selection and live telemetry projection.
FR-14.1 — Normal flow	Display authorized live position, altitude with reference, speed, battery percentage, connectivity, source/received timestamps, freshness and mission progress. Progress is the count of confirmed reached planned waypoints divided by the execution snapshot waypoint count, expressed as a percentage; absent reliable waypoint progress is Unavailable. Live state updates as accepted samples arrive and links to related incidents.
FR-14.2 — Validation / Alternative flow	The system shall enforce the following checks and outcomes: No packet/stale stream shows last-known values with timestamp and stale/offline status; never imply stale values are live; reconnect refreshes the latest valid state.
FR-14.3 — Postcondition	The system shall provide: Real-time operational view with explicit freshness.
Business rules	BR-07; BR-08
Traceability	UC-14

Screen layout — SCR-14



SCR-10 — Screen layout.

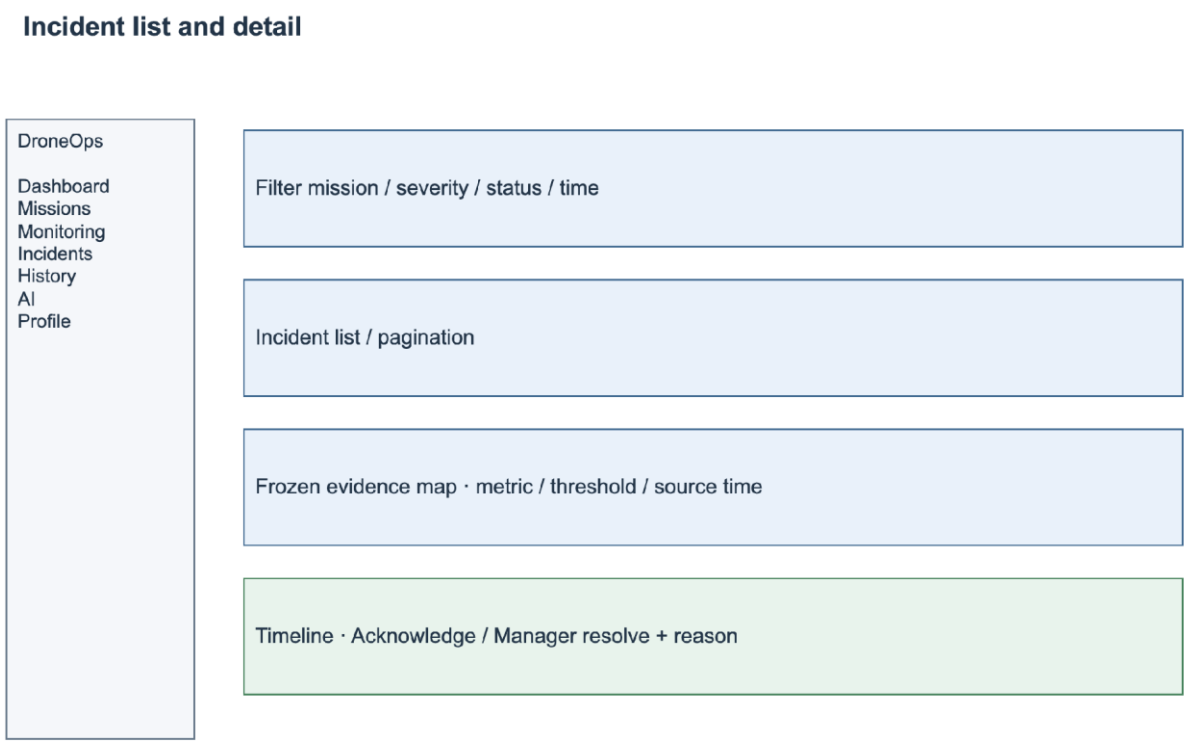
3.8 FE-07 — Operational Alert and Incident Detection

3.8.1 Detect incidents and deliver alerts (UC-15)

Specification	Details
Actors / Interface	UAV/Simulator (data source); Operator; Manager | SCR-14 / SCR-15 / SCR-26 / background
Trigger	Accepted telemetry or connectivity-check timer.

Specification	Details
Preconditions	A configured source/UAV mapping and enabled detection rules exist. Background evaluation uses an authorized integration identity and does not require a human session.
Input / Data	Telemetry, last-seen time and configured abnormal-event rules.
FR-15.1 — Normal flow	The system shall support the following behavior: Evaluate communication loss, low battery and abnormal altitude/speed; create an incident linked to UAV/mission and evidence; persist it and deliver an in-app alert to permitted users. Run communication-loss checks even with no new packet.
FR-15.2 — Validation / Alternative flow	Missing measurements disable only the affected rule and expose Unavailable. Deduplicate incidents by UAV, rule and abnormal episode, including concurrent timer/packet evaluation. A delivery failure retains the incident and unread in-app alert for subsequent retrieval. Recovery follows BR-15; acknowledgement or resolution cannot manufacture a recovery sample.
FR-15.3 — Postcondition	The system shall provide: Persisted incident and available in-app alert; no autonomous control action.
Business rules	BR-08; BR-09
Traceability	UC-15

Screen layout — SCR-14 / SCR-15 / SCR-26



SCR-11 — Screen layout.

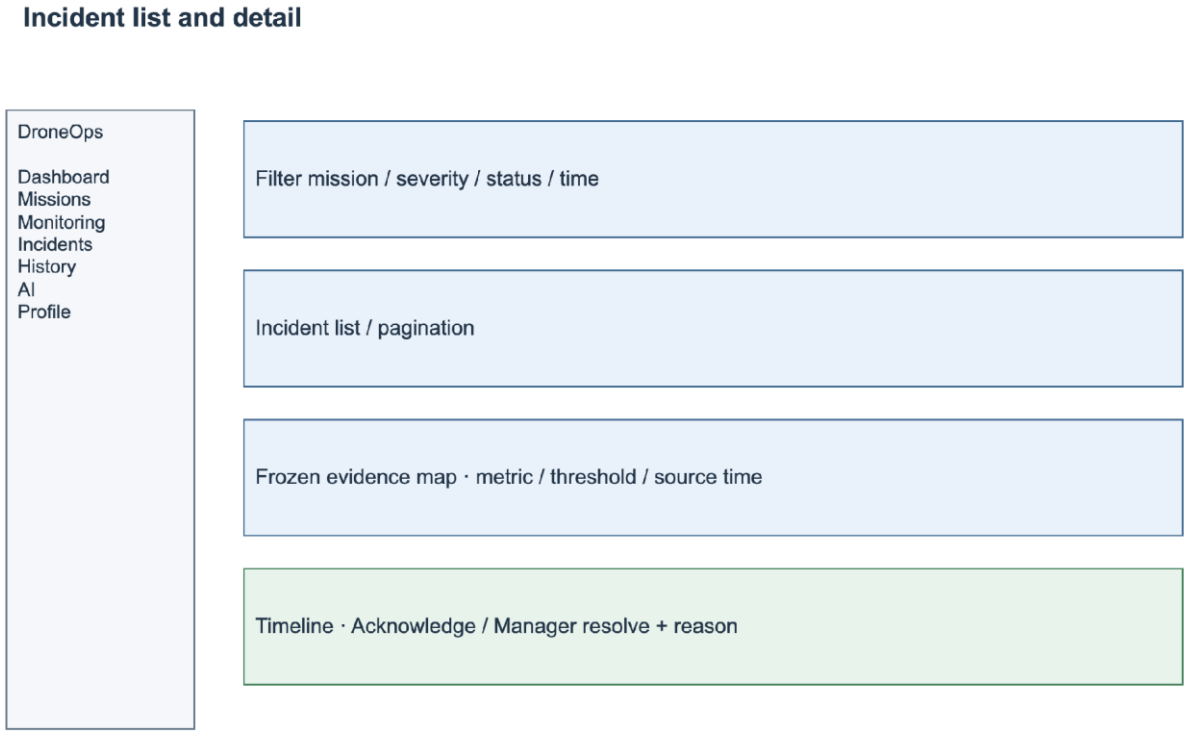
Detection rules use the configured thresholds and recovery conditions in BR-15. One continuous abnormal episode produces one incident per UAV, rule and flight context. Acknowledgement and manual resolution do not reset the detector; only measured recovery ends the episode.

3.9 FE-08 — Incident Management and Post-Flight Analysis

3.9.1 Review and manage an incident (UC-16)

Specification	Details
Actors / Interface	Operator; Manager | SCR-15 / SCR-16 / SCR-26
Trigger	Incidents > select incident; acknowledge or resolve if permitted.
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Incident ID, linked evidence and acknowledgement/resolution note.
FR-16.1 — Normal flow	The system shall support the following behavior: View time, type, affected mission/UAV and telemetry evidence. The workflow allows assigned operator/manager acknowledgement and manager resolution, with actor/time and notes retained.
FR-16.2 — Validation / Alternative flow	The system shall enforce the following checks and outcomes: Unauthorized transition rejected; concurrent update requires reload; absent evidence is labelled unavailable. Acknowledgement does not alter measured telemetry or suppress unrelated alerts.
FR-16.3 — Postcondition	The system shall provide: Reviewed incident with append-only status history.
Business rules	BR-01; BR-09; BR-12
Traceability	UC-16

Screen layout — SCR-15 / SCR-16 / SCR-26



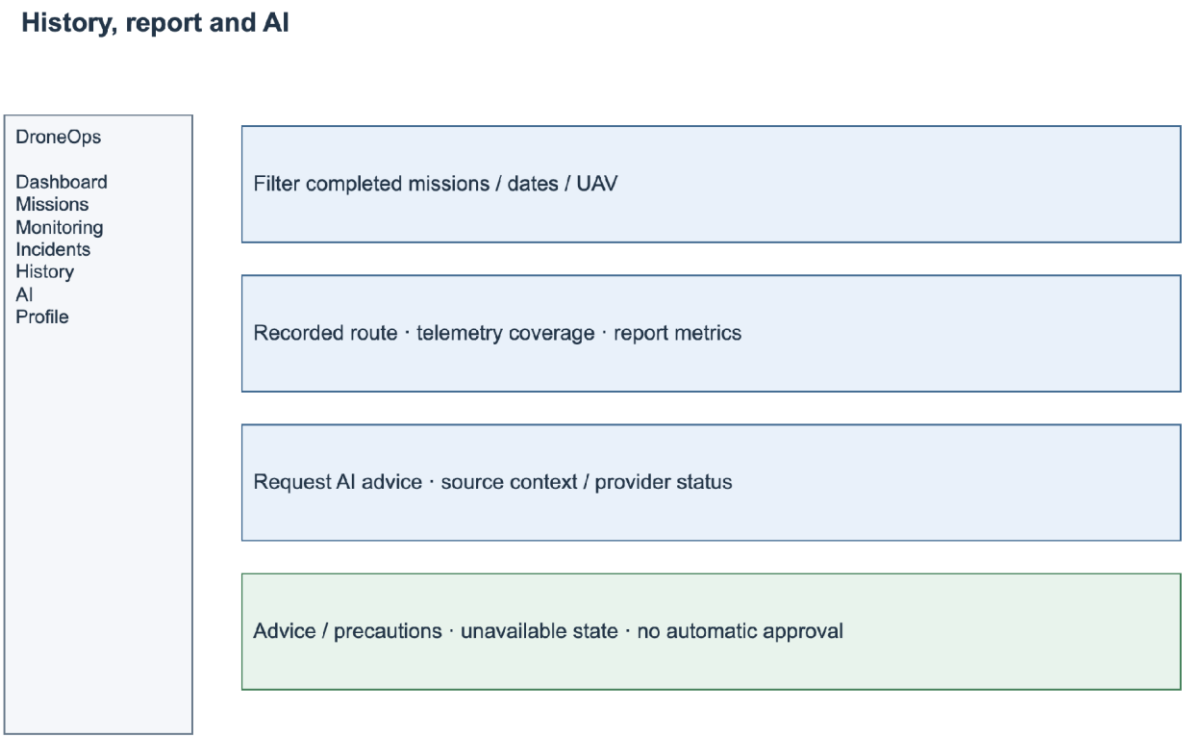
SCR-12 — Screen layout.

3.9.2 Finalize a flight and generate a post-flight report (UC-17)

Specification	Details
Actors / Interface	UAV/Autopilot or Simulator (completion source) | SCR-18 / SCR-27 / background
Trigger	Configured completion event accepted for a known mission/flight.

Specification	Details
Preconditions	An authorized completion event identifies the mission and its persisted execution data.
Input / Data	Flight identity, completion state/time, stored telemetry and incidents.
FR-17.1 — Normal flow	Recognize a configured, trusted mission-complete event only for the mapped active flight; armed/disarmed state or communication loss alone is insufficient. Atomically finalize the flight and mission as Completed, capture completion time and produce a report from retained execution, telemetry and incident evidence. Persist report status Queued, Generating, Available or Failed and its data coverage window.
FR-17.2 — Validation / Alternative flow	Ambiguous or duplicate completion events cannot create duplicate flights/reports. Failed generation preserves all evidence and exposes a failure reason. Managers may retry generation for completed flights; Operators may view status. Late telemetry is retained and marks the available report Outdated when it falls within that flight window. A Manager regeneration creates a new report revision while preserving previous reports.
FR-17.3 — Postcondition	The system shall provide: Completed flight history and available or explicitly failed report.
Business rules	BR-07; BR-10
Traceability	UC-17

Screen layout — SCR-18 / SCR-27



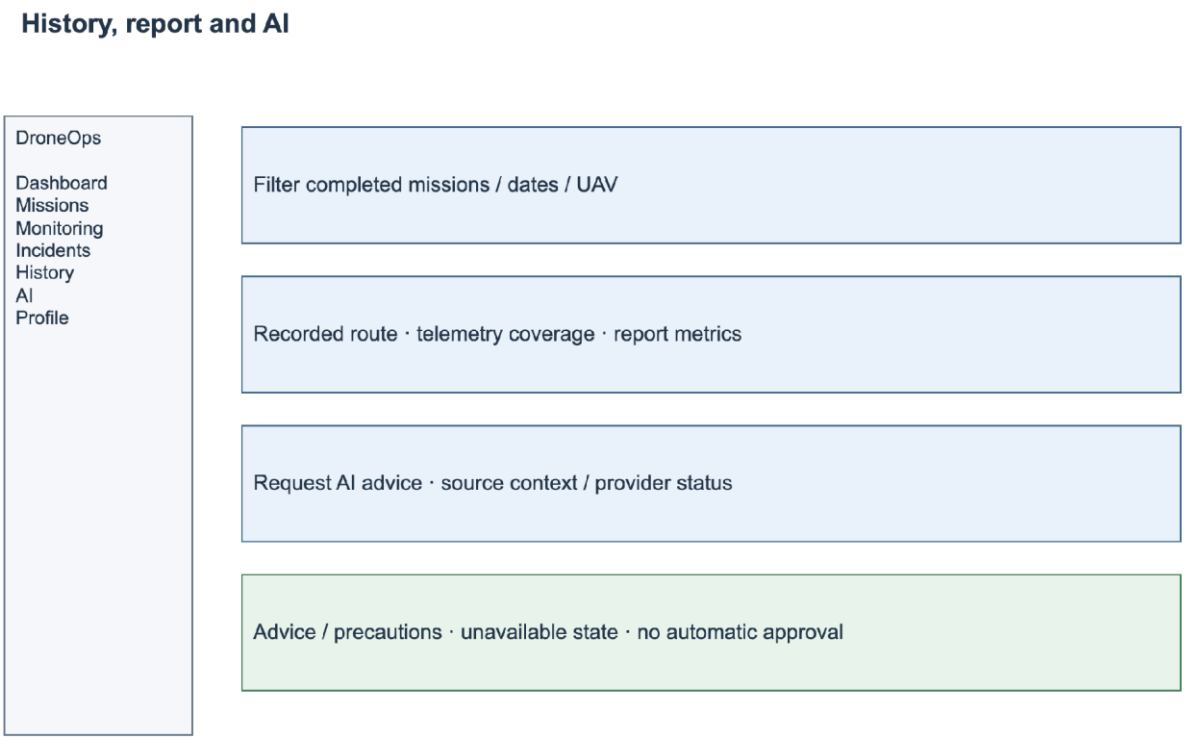
SCR-14 — Screen layout.

3.9.3 Review mission history and post-flight results (UC-18)

Specification	Details
Actors / Interface	Operator; Manager | SCR-17 / SCR-18 / SCR-27
Trigger	Flight History > select flight > Report / Incidents.
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Date/status/UAV filters and flight ID.

Specification	Details
FR-18.1 — Normal flow	The system shall support the following behavior: List permitted histories; show planned-versus-recorded route where data exists, duration/telemetry summaries, incidents and post-flight report; link incident details.
FR-18.2 — Validation / Alternative flow	Show a scoped empty state when no records match. Label missing or incomplete telemetry and display report coverage. Queued/Generating/Failed/Outdated reports show their status; only Managers may retry or regenerate. This release provides on-screen report viewing; downloadable report export is outside scope.
FR-18.3 — Postcondition	The system shall provide: User reviews recorded flight evidence and report.
Business rules	BR-01; BR-10
Traceability	UC-18

Screen layout — SCR-17 / SCR-18 / SCR-27



SCR-13 — Screen layout.

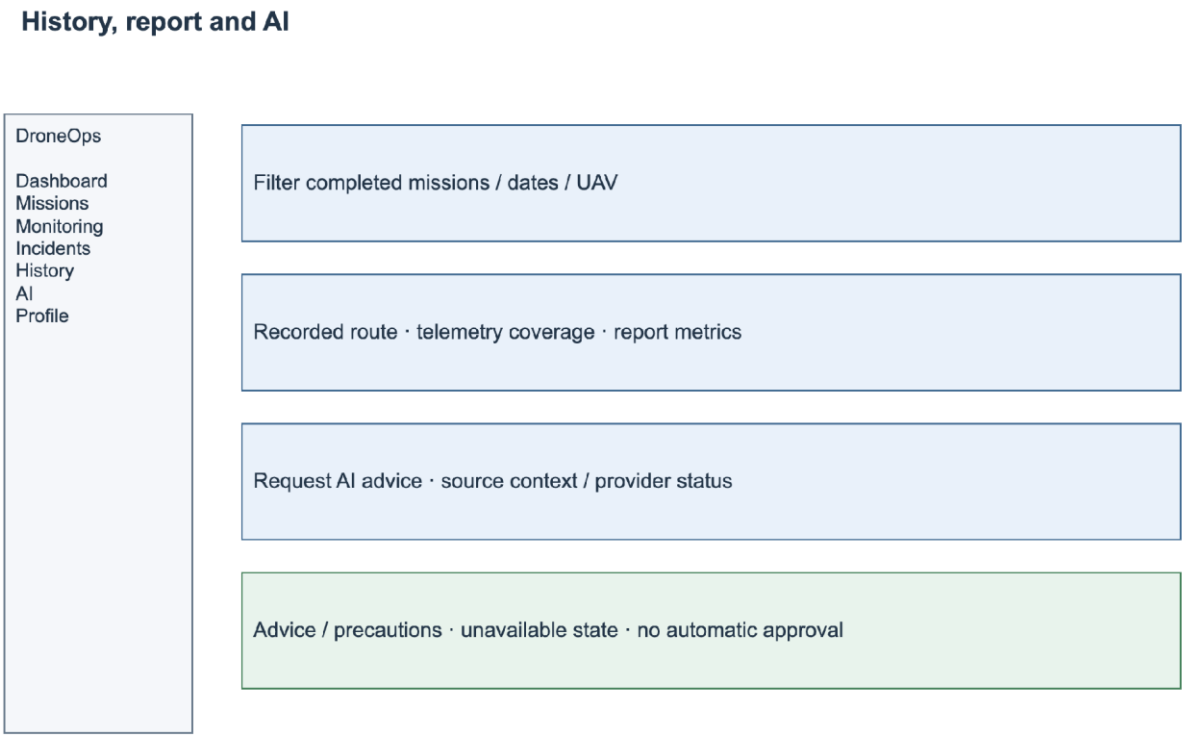
3.10 FE-09 — AI-Assisted Mission Decision Support

3.10.1 Request an AI-assisted risk assessment (UC-19)

Specification	Details
Actors / Interface	Operator; Manager | SCR-19
Trigger	New / Detail / Edit Mission or Live Monitoring > AI Risk Assessment > Generate.
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Selected current mission or proposed update payload, validation results and available telemetry/incidents.

Specification	Details
FR-19.1 — Normal flow	Authorize the mission or pending proposal, capture its context and validation/telemetry availability, and retrieve up to five similar completed flights using normalized route, duration, altitude, vehicle and incident features. Retain source record identifiers and similarity scores. A configured OpenAI or Gemini adapter may summarize the retrieved evidence into mission summary, potential risks and precautions; preserve source links and distinguish observed historical evidence from generated advice.
FR-19.2 — Validation / Alternative flow	Missing essential plan data blocks assessment. No historical matches produces an explicit No historical evidence result, not a zero-risk conclusion. Timeout, rate limiting or invalid provider output shows MSG-11 while preserving retrieved evidence and manual review. Never present generated or fabricated missions as historical evidence. Credentials remain server-side.
FR-19.3 — Postcondition	The system shall provide: Available assessment or explicit failed status; no change to mission approval.
Business rules	BR-11; BR-12
Traceability	UC-19

Screen layout — SCR-19



SCR-15 — Screen layout.

3.10.2 Review AI advice and supplied context (UC-20)

Specification	Details
Actors / Interface	Operator; Manager | SCR-19
Trigger	Open a completed assessment.
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Assessment ID, source mission/update request ID, captured input/time and provider result.

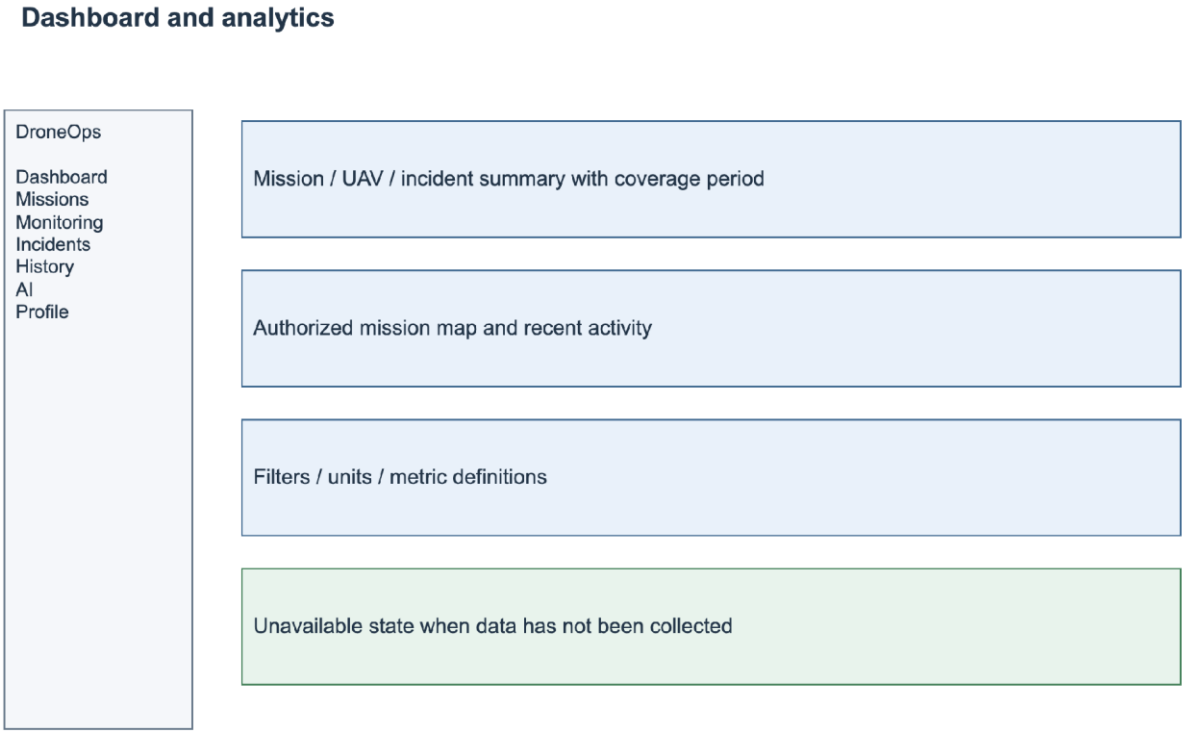
Specification	Details
FR-20.1 — Normal flow	Display the assessed mission/proposal, request time, retrieval method, returned historical flights, similarity scores, relevant characteristics and linked incidents alongside summary, potential risks and precautions. Clearly label generated advice and missing evidence. Both roles may request a new assessment; mission edits follow UC-06.
FR-20.2 — Validation / Alternative flow	The system shall enforce the following checks and outcomes: Changed mission/context marks the result outdated; absent data is shown; users cannot overwrite stored provider output and present it as the original result. No model confidence is invented.
FR-20.3 — Postcondition	The system shall provide: Human-reviewed advisory evidence, with manual mission review still available.
Business rules	BR-06; BR-11
Traceability	UC-20

3.11 FE-10 — Mission and Operational Dashboard

3.11.1 View an operational dashboard (UC-21)

Specification	Details
Actors / Interface	Operator; Manager | SCR-05
Trigger	Sign in > Dashboard.
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Authorized scope and date/status/UAV filters.
FR-21.1 — Normal flow	The system shall support the following behavior: Show mission counts/status, active UAV summaries, recent incidents and shortcuts to validation, monitoring, history and AI assessment; drill down to authorized records. Both roles can change view filters.
FR-21.2 — Validation / Alternative flow	The system shall enforce the following checks and outcomes: No data or partial service failure displays scoped empty/unavailable states; never expose inaccessible counts/details. Dashboard metrics do not directly edit mission evidence.
FR-21.3 — Postcondition	The system shall provide: Role-scoped operational overview.
Business rules	BR-01; BR-07
Traceability	UC-21

Screen layout — SCR-05



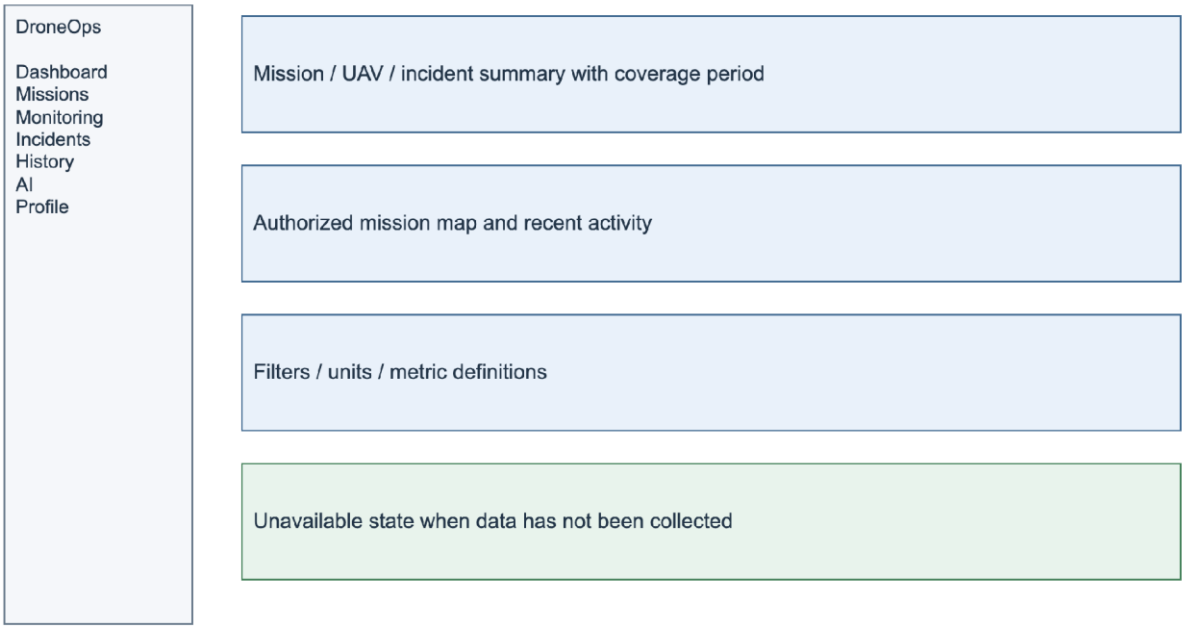
SCR-02 — Screen layout.

3.11.2 Review operational summaries (UC-22)

Specification	Details
Actors / Interface	Operator; Manager | SCR-05
Trigger	Operational Dashboard > authorized history and summary links.
Preconditions	The user is authenticated and has permission to access the selected mission or record.
Input / Data	Permitted date range, UAV/mission filters and recorded histories/incidents.
FR-22.1 — Normal flow	Show completion rate, flight duration, recorded distance, incident counts by type/severity and trends within the authorized date/UAV scope. Calculate completion rate as Completed flights divided by all started flights in the selected start-time window, with an empty denominator shown as Unavailable. Duration uses trusted flight start/end timestamps. Recorded distance sums consecutive valid GPS positions and excludes gaps longer than the communication-loss threshold, reporting excluded coverage. Counts use distinct incident IDs, not status events. Display units, window and source links.
FR-22.2 — Validation / Alternative flow	Invalid date ranges shall be rejected. Missing data shall be displayed as unavailable and shall not be interpreted as zero risk.
FR-22.3 — Postcondition	The system shall provide: Explainable aggregate view backed by recorded data.
Business rules	BR-01; BR-10
Traceability	UC-22

Screen layout — SCR-05

Dashboard and analytics



SCR-16 — Screen layout.

4. Non-Functional Requirements

4.1 External Interfaces

ID	Interface	Required behavior / Contract boundary
IF-01	Human Web interface	Web screens shall provide role-scoped navigation, visible validation and status feedback, and explicit units and timestamps. Textual findings shall remain available when the map cannot load.
IF-02	Operator mobile client	The Operator mobile client supports sign-in/out, own profile, own/assigned mission and validation/decision viewing, live monitoring, alerts and flight history under the same authorization rules. Route authoring and Manager administration are Web-only.
IF-03	UAV/autopilot or Simulator -> gateway	The configured gateway receives MAVLink 2 telemetry over UDP or TCP and bridges normalized messages to platform services over authenticated WebSocket or MQTT. Retain source/UAV identity, session/sequence, source and received timestamps, units and altitude reference. Enforce configured source mappings; unsupported fields remain unavailable.
IF-04	Gateway -> platform -> live clients	The telemetry gateway shall normalize and persist messages, preserve source and received timestamps, deduplicate replays and publish authorized live-state updates.
IF-05	Backend -> external AI provider	Historical-mission retrieval returns scoped source IDs, feature/similarity evidence and related incidents. The optional external AI adapter receives bounded authorized context and returns structured summary, risks and precautions. Provider/model, context limit and timeout are configurable; default request timeout is 30 seconds.
IF-06	Map/background data	The map interface renders WGS84 route and restriction geometry, supports waypoint/polygon selection and provides attribution required by the selected provider. Provider failure leaves coordinate entry and textual validation available. Restrictions are entered through the polygon editor in this release; external file import is outside scope.
IF-07	Persistent storage	The system shall retain current missions, update requests, review decisions and linked telemetry, incident, report and assessment evidence.

Performance profile P-01: 5 simulated UAVs at 1 normalized sample/second each; 10 concurrent Web sessions; 100 stored flights with 10,000 samples each; and up to 50 waypoints checked against 20 configured polygons, each containing at most 100 vertices. External map and AI response times are measured separately.

4.2 Quality Attributes

4.2.1 Usability

ID	Requirement	Verification
NFR-U01	After at most 30 minutes of onboarding, at least 4 of 5 representative testers shall create, validate and submit a prepared 5-waypoint mission within 5 minutes without assistance.	Timed task study.
NFR-U02	All required-field errors, stale-data states and permission-denied responses shall be understandable without consulting server logs. Core forms shall be usable with keyboard focus and text labels.	Screen walkthrough and keyboard/error-state checks.

4.2.2 Reliability

ID	Requirement	Verification
NFR-R01	Under P-01, a 60-minute monitoring run shall have no application crash or loss of acknowledged persisted mission/decision/incident records.	Controlled soak test and database/trace evidence.

ID	Requirement	Verification
NFR-R02	A disconnected stream shall be marked stale/offline within the configured communication-loss threshold plus 2 seconds; restored data shall appear within 5 seconds of accepted reconnect data.	Timed disconnect/reconnect tests using BR-15 thresholds under P-01.
NFR-R03	AI timeout/provider failure shall not prevent authorized users from opening mission details and performing manual review/decision actions.	Inject provider timeout/rate-limit/invalid response; verify UC-09 remains available.
NFR-R04	A failed report-generation attempt shall preserve flight evidence, expose Failed status and allow scoped Manager retry for completed flights.	Inject generation failure, retry and late-data regeneration; verify retained revisions.

4.2.3 Performance

ID	Requirement	Verification
NFR-P01	Under P-01, 95% of normal list/detail requests shall complete within 2 seconds, excluding external map/AI latency.	Measure at least 100 requests after warm-up; document hardware/network.
NFR-P02	Under P-01, 95% of accepted telemetry samples shall reach the authorized live view within 2 seconds of gateway receipt.	Correlate gateway-received and client-rendered timestamps with synchronized clocks.
NFR-P03	Under the P-01 route/restriction limits, 95% of pre-flight validations shall finish within 3 seconds.	At least 30 controlled valid/invalid route cases.
NFR-P04	The AI request timeout shall be configurable. When it expires, the application shall leave the loading state and provide retry and manual-review options.	Delayed-provider response test with application timeout verification.

4.2.4 Security and Privacy

ID	Requirement	Verification
NFR-S01	Every protected screen operation and corresponding API shall reject unauthorized role or out-of-scope record access, including direct identifier requests.	Positive/negative matrix tests for section 3.1.3.
NFR-S02	Passwords shall be stored as secure hashes. Provider credentials shall remain on the server. Authentication and application data shall use encrypted transport.	Configuration/code review and response/log inspection.
NFR-S03	AI requests shall include only authorized relevant mission context and shall exclude credentials and unnecessary account identifiers.	Inspect authorized request payloads and configured retention policies.
NFR-S04	License images shall be private and accessible only to the owner and authorized reviewers through expiring access. Reject unsupported image types or files above 5 MB. Do not persist credentials, OTP values or private object access tokens in application logs.	Cross-account access tests, expired-link checks and upload validation.
NFR-S05	Retain mission decisions, execution evidence, telemetry, incidents and reports for 12 months after flight completion. Retain superseded/rejected license images for 90 days after the decision; retain the current image while the account is active. Authorized retention jobs remove expired data without leaving broken active references and audit the deletion.	Time-controlled retention tests including dependent-record and media cleanup.

4.2.5 Maintainability and Testability

ID	Requirement	Verification
NFR-M01	FE/UC/FR identifiers shall map to implementation and test cases for release acceptance. Core rules, timers, retrieval and external adapters shall support deterministic fixtures.	Traceability review and repeatable fixture-based test runs.
NFR-M02	Evaluate historical retrieval on a fixed labeled mission set with documented relevance criteria, feature configuration and k=5. Report Precision@5, returned evidence links and query latency; manually assess fewer-than-five and no-match cases.	Retain labeled queries, expected relevant records and reproducible measured results.

4.2.6 Compatibility and Recovery

ID	Requirement	Verification
NFR-C01	The Web application shall support Chrome and Edge at desktop viewport widths of 1280 and 1440 pixels.	Record exact tested versions at release; responsive screen review.
NFR-C02	The system shall support scheduled backups and restoration of persisted records and linked private media according to NFR-C03.	Backup/restore rehearsal.
NFR-C03	Daily backups shall cover mission, review, account/license metadata, telemetry and incident records. Recovery point objective is 24 hours and recovery time objective is 4 hours for the reference deployment. Storage-object recovery shall be included in the rehearsal.	Restore a backup and linked media in an isolated environment; measure data gap and elapsed time.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-01	Every human request is authenticated and authorized for the action and record scope; external telemetry/AI integrations use configured integration credentials, not human login.
BR-02	UAVs and pilots must be active and eligible. A pilot license must be approved, unrevoked and valid through the scheduled flight end. Deactivation preserves history and is blocked for an active flight. Assignment conflicts follow BR-14.
BR-03	Approval applies only to the reviewed mission or proposed update after mandatory spatial checks pass with current evidence. Applying a proposal updates the same mission atomically. Incomplete/error/stale validation cannot authorize approval.
BR-04	Both waypoints and connecting route segments shall be validated against active restrictions using their configured coordinate units, altitude reference and boundary rules.
BR-05	Initial mission submission and every later pre-flight edit require Manager review. Initial Draft creation is not submission. No application action issues low-level flight-control commands. Lifecycle rules follow BR-13.
BR-06	Mission changes, validation runs and decisions link to mission ID and update request ID. Every accepted edit requires Manager review. Approved current content and status remain unchanged until a proposal is approved; rejected proposals never overwrite them. Earlier evidence cannot authorize changed content.
BR-07	Telemetry and displayed summaries retain source/received timestamps and missing-data indicators. Unknown or stale measurements must not appear as fresh valid data.
BR-08	Communication loss shall be evaluated by an independent timer as well as on packet arrival. Timeout and recovery conditions shall be configurable.
BR-09	Incidents shall retain type, time, UAV/mission association and frozen event evidence. Acknowledgement and resolution shall record actor, timestamp and reason without rewriting source telemetry.
BR-10	Post-flight reports and analytics shall use stored mission, telemetry and incident evidence and show missing data and coverage explicitly. Calculated metrics shall retain their units and covered time window.
BR-11	AI output provides mission summaries, risk observations and precautions for human review. It does not approve/reject a mission or command a UAV. API failure must not block manual review.
BR-12	Record actor/integration identity, action, target and timestamp for material changes; protect credentials and restrict source context sent to AI to permitted mission data.
BR-13	Mission lifecycle: Draft -> Pending Review -> Approved or Rejected. Draft/Rejected may be initially submitted; each accepted edit to a non-executing mission creates a pending update without changing current content/status. Update approval applies content and sets Approved; update rejection preserves current content/status. A newer accepted proposal supersedes the previous pending review. Draft/Pending Review/Rejected/Approved may be cancelled; executing or terminal missions cannot be edited. Flight start requires Approved, fresh validation, eligible assignments and no pending update. A trusted start event creates one flight and moves the mission to In Progress; a trusted completion event moves it to Completed. Reflight requires a new mission.
BR-14	Schedules use UTC instants and display the user time zone. End must exceed start. Approved missions reserve UAV/pilot intervals [start,end); adjacent intervals do not conflict. Pending submissions/proposals are checked again atomically at approval and flight start. A proposal cannot reserve over another approved booking; its existing approved reservation is replaced only on approval.
BR-15	Default detector settings: communication loss after 10 seconds without a valid mapped sample, checked at least once per second; recovery requires two consecutive valid samples. Battery below 20% for two consecutive samples is Warning; recovery requires two samples at or above 25%. Altitude outside the approved/configured MSL range and speed above the configured UAV maximum for two consecutive samples are Critical; recovery requires two samples within limits. Communication loss is Critical. Missing thresholds or altitude conversion disable the affected rule with visible diagnostics. Manager-configured changes are audited; severity escalations update the same incident with retained evidence.

ID	Rule Definition
BR-16	The author cannot approve their own proposed mission update or license-change request. Another authorized Manager must review it. No mission may start with a pending proposal; no flight-control action is issued to enforce this planning decision.
BR-17	Keep current account email and approved license until replacement verification/review succeeds. A rejected or cancelled license request does not modify current credentials. Expiry of the current license still affects eligibility while a replacement is pending.
BR-18	Historical similarity retrieval uses up to five completed flights in the authorized data scope. Normalize numeric features with a recorded feature schema and score method; display fewer than five when fewer eligible records exist. Store feature/index configuration and evidence IDs with each assessment. AI provider output cannot change approval or detector thresholds.

5.2 Common Requirements

ID	Requirement
CR-01	Enforce authentication/action/record authorization at the backend; session expiry returns the user to sign-in without treating an unsaved change as persisted.
CR-02	Validate required fields, ranges and unique identifiers on the server and provide field-level feedback. Do not rely solely on client validation.
CR-03	Persist timestamps in UTC and display them with an explicit user-zone label; default display zone is Asia/Ho_Chi_Minh. Display metres MSL for planned altitude, m/s for speed, metres/kilometres for distance and percent for battery. Preserve incoming altitude reference; unavailable conversion is explicit.
CR-04	Search/filter/pagination must preserve permitted scope and show explicit loading, empty, error and stale states; default page size is a UI decision.
CR-05	Prevent silent concurrent overwrite using request identity, a current-payload fingerprint and an atomic pending-request pointer, without mission versioning. Only the current pending request may be decided. A stale form returns a conflict; retry of the same operation does not duplicate a review, decision, flight, alert or incident. Preserve the approved execution snapshot independently of later source changes.
CR-06	Confirm destructive/deactivation/cancellation actions; preserve history required by references and retention policy. No user-facing Cancel action issues a UAV stop command.
CR-07	Changes and material automated outcomes retain actor/source, target, time and result evidence; do not expose credentials in errors or audit output.
CR-08	Use consistent English application messages; escape substituted values before display. Show field-level failures and prominent success/error feedback with keyboard-accessible status text. Changing presentation language does not alter persisted evidence.

5.3 Application Messages List

Code	Type	Context	Content
MSG-01	Inline	Required/invalid field	Enter a valid {field}.
MSG-02	Inline	Invalid login	Unable to sign in. Check your credentials.
MSG-03	Page / toast	Access denied	You do not have permission to perform this action.
MSG-04	Toast	Save succeeds	Changes saved successfully.
MSG-05	Dialog	Concurrent update	This record has changed. Reload before saving.
MSG-06	Banner	Mandatory validation failure	Mission validation failed. Review the highlighted violations.
MSG-07	Banner	Validation incomplete	Validation is unavailable or incomplete. Approval is blocked.
MSG-08	Toast	Submitted	Mission submitted for review.

Code	Type	Context	Content
MSG-09	Banner	Stale telemetry	Telemetry is stale. Last update: {timestamp}.
MSG-10	In-app alert	Incident raised	{incidentType} detected for {uavName}. Open incident details.
MSG-11	Banner	AI unavailable	AI assessment is unavailable. Retry later or continue manual review.
MSG-12	Banner	Outdated assessment	Mission data has changed since this assessment. Generate a new assessment.
MSG-13	Empty state	No matching data	No matching records found.
MSG-14	Report status	Generation failed	The report could not be generated. Flight data is preserved.
MSG-15	Confirmation	Cancel plan	Cancel this mission plan? This does not stop or control a UAV.
MSG-16	Inline	Rejection reason absent	Enter a reason for rejecting this mission.
MSG-17	Banner	Map unavailable	The map is unavailable. Review the listed coordinates and validation results.
MSG-18	Banner	Session expired	Your session has expired. Sign in again.
MSG-19	Inline / banner	Email already registered	This email is already registered. Enter another email.
MSG-20	Success banner	Verification sent	A verification code was sent to {email}. Check your inbox.
MSG-21	Inline	Invalid or expired code	The verification code is invalid or expired. Request a new code.
MSG-22	Success banner	Email changed	Your email has been changed. Sign in again.
MSG-23	Banner	License request submitted	Your license change is awaiting review. Your current license remains unchanged.
MSG-24	Banner	License decision	Your license change was {decision}. {reason}
MSG-25	Banner	Pending mission update	An update is awaiting review. This mission cannot start until review is complete.
MSG-26	Banner	No historical evidence	No similar historical flights are available. Continue manual review.
MSG-27	Inline	Verification rate limit	Please wait {seconds} seconds before requesting another code.

5.4 Requirements Traceability

Feature	Use Cases	Detailed FRs	Screens / Non-screen
FE-01	UC-01–UC-04; UC-23–UC-26	FR-01.1–FR-04.3; FR-23.1–FR-26.3	SCR-01; SCR-03–SCR-05; SCR-17–SCR-21; NSF-09/10
FE-02	UC-05, UC-06, UC-07	FR-05.1-.3; FR-06.1-.3; FR-07.1-.3; FR-AF-01/04	SCR-07; SCR-06 / SCR-08
FE-03	UC-08, UC-09, UC-10	FR-08.1-.3; FR-09.1-.3; FR-10.1-.3; FR-AF-01	SCR-08; SCR-09
FE-04	UC-11, UC-12	FR-11.1-.3; FR-12.1-.3	SCR-08 / background; SCR-08
FE-05	UC-13	FR-13.1-.3; FR-AF-03/04	Non-screen
FE-06	UC-14	FR-14.1-.3; FR-AF-02/03	SCR-10
FE-07	UC-15	FR-15.1-.3	SCR-10 / SCR-11 / background
FE-08	UC-16, UC-17, UC-18	FR-16.1-.3; FR-17.1-.3; FR-18.1-.3	SCR-11 / SCR-12; SCR-14 / background; SCR-13 / SCR-14

Feature	Use Cases	Detailed FRs	Screens / Non-screen
FE-09	UC-19, UC-20	FR-19.1-.3; FR-20.1-.3	SCR-15; NSF-07/11
FE-10	UC-21, UC-22	FR-21.1-.3; FR-22.1-.3	SCR-02; SCR-16

5.5 Terms and Data Conventions

Mission: the current planning record with one stable ID. Mission update request: a separately stored proposed change requiring review, not an executable mission or mission version. Flight: the immutable execution record created from an approved mission at start.

MSL: altitude above mean sea level. AGL: altitude above local ground; conversion requires terrain/reference data. WGS84: geographic coordinate reference for latitude and longitude.

Fresh telemetry: valid mapped data received within the configured communication-loss threshold. Stale/offline: the threshold has elapsed; last-known values remain visible with timestamps. An incident episode persists until measured recovery.

Top-k retrieval: selecting up to k relevant completed flights using a recorded feature representation and similarity method. Similarity is not a calibrated probability of safety or incident occurrence.

Automatic flight preparation and workflow ownership

Current ownership: MF1 / WF-01 — NghiaHQ (Ha Quang Nghia), planning and preflight; MF2 / WF-02 — ThongLQ (Le Quoc Thong), realtime monitoring, incidents and flight preparation; MF3 / WF-03 — DucDDA (Dang Dao Anh Duc), post-flight; MF4 / WF-04 — KhanhGNG (Giang Nguyen Gia Khanh), AI assessment. ThongLQ remains team leader and dossier coordinator.

Implemented scope

AutomaticFlightRequested is optional and defaults to false. Changes follow the mission proposal and approval workflow. This release prepares readiness; dispatchEnabled is false. Neither confirmation nor gateway enrollment uploads a mission, arms motors, sends TAKEOFF/AUTO/MISSION_START, or changes the mission to InProgress.

Confirmation conditions

For an approved, opted-in mission with a schedule, the backend opens confirmation at T minus 5 minutes and closes it at the scheduled start. The authenticated operator must keep a visible browser tab online. Presence is sent every 10 seconds and expires after 30 seconds. The selected, unrevoked mission connection must match the active UAV and flight-controller UID, with gateway heartbeat within 30 seconds and telemetry within 15 seconds.

Readiness evidence

Confirmation reruns preflight and binds readiness to the exact plan hash and connection ID. Readiness requires a passing preflight for that hash from the last minute. Changed plan/connection, stale presence or missing telemetry invalidates the conditions; the operator must review and confirm again. Dismissal, a hidden/offline tab or a missed window is not flight consent.

Future execution scope

Autopilot execution remains a separate development phase: firmware/airframe capability detection, mission compilation and upload/read-back, command IDs/deadlines and replay protection, scheduled condition rechecks, local pre-arm/failsafe enforcement, and ACK plus observed flight-state verification. Validate ArduPilot/PX4 SITL, propeller-free bench operation and supervised flight before enabling dispatch. AI advice cannot authorize launch.

FR-AF-01 Opt in and review

Actor: mission owner; reviewer: authorized Manager. Saving a changed automatic-flight option creates reviewed proposed content. Acceptance: the option defaults off, unapproved changes do not replace approved content, and readiness is unavailable for a non-opted-in or non-approved mission.

FR-AF-02 Confirm readiness

Actor: authenticated mission owner with uav_operator role. Acceptance: only T-5 minutes <= backend now < start permits confirmation; another owner cannot access the mission; missing/stale presence, heartbeat, telemetry or preflight blocks readiness. Success stores the current plan hash and connection, without sending aircraft commands.

FR-AF-03 Cancel and invalidate

Acceptance: cancelling clears confirmation and presence. A changed hash/connection or expired presence must not reuse earlier confirmation. A missed start window requires rescheduling and review; reconnection must not imply late launch consent.

FR-AF-04 Export plan

GET /api/missions/{id}/automation/plan returns an authorized plan envelope, including mission/UAV IDs, planHash, UTC schedule, speed in m/s, altitude reference and ordered waypoints with WGS84 degrees/E7 and altitude metres. dispatchEnabled remains false; this is not a compiled MAVLink mission.